Check for updates

Received: 13 August 2026 | Revised: 15 September 2026 | Accepted: 23 September 2026

DOI: 10.1002/ep.70734

ENVIRONMENTAL PROGRESS & SUSTAINABLE ENERGY

ORIGINAL RESEARCH

Sustainable Energy

# Topology-porosity assessment of TPMS-based aluminum foams for accelerated melting of RT31 phase change material

Sirine Chtourou[1] | Ines Ben Ali[2] | Lina Chelbi[3] | Mounir Baccar[4]

[1]Mechanical Engineering Department, National School of Engineers of Tunis (ENIT), University of Tunis El Manar, Tunis, Tunisia, Laboratory of Advanced Fluid Dynamics, Energy and Environment, Sfax, Tunisia

[2]National School of Engineers of Tunis (ENIT) and ENSTA Paris, Polytechnic Institute of Paris, Paris, France

[3]Mechanical Engineering at National School of Engineers of Sfax (ENIS), Laboratory Advanced Fluid Dynamics, Energetic and Environment, National School of Engineers of Sfax (ENIS), University of Sfax, Sfax, Tunisia

[4]The Mechanical Department of the National School of Engineers of Sfax (ENIS), Laboratory Advanced Fluid Dynamics, Energetic and Environment, National School of Engineers of Sfax (ENIS), University of Sfax, Sfax, Tunisia

**Correspondence**
Sirine Chtourou, The Mechanical Department of the National School of Engineers of Tunis (ENIT), University of Tunis El Manar, Tunis; Laboratory Advanced Fluid Dynamics, Energetic and Environment, The National School of Engineers of Sfax (ENIS), University of Sfax, Road Soukra km 3.5, Sfax 3038, Tunisia.
Email: sirine.chtourou@enis.tn

## Abstract

Thermal energy storage systems based on phase change materials are limited by the low thermal conductivity of PCMs, which results in slow melting and temperature non-uniformity. In this study, a three-dimensional numerical investigation was performed to assess the combined effects of TPMS (Triply Periodic Minimal Surface) topology and porosity on the melting behavior of RT31 PCM. Gyroid, Schwarz Diamond, and Schwarz Primitive structures were examined using the enthalpy-porosity method in ANSYS Fluent R19. The numerical model showed good agreement with the experimental RT31 data, with an RMSE of 1.24°C and an $R^2$ of 0.995. At the common porosity of 85.11%, the Gyroid configuration reached 95% melting in 57.29 s, compared with 100.89 s for Schwarz Diamond and 99.91 s for Schwarz Primitive, corresponding to reductions of 43.2% and 42.7%, respectively. For the Gyroid topology, decreasing porosity from 92.03% to 78.18% reduced the time to 95% melting from 65.76 to 50.32 s, an improvement of approximately 23.5%. However, lower porosity also reduced PCM volume and restricted natural convection. The results therefore show that intermediate Gyroid porosities provide the most balanced compromise between rapid melting, thermal uniformity, natural-convection development, and latent storage capacity.

**KEYWORDS**
ANSYS Fluent R19, gyroid, melting time, phase change materials, porosity, thermal energy storage, TPMS

## 1 | INTRODUCTION

The worldwide transition toward sustainable energy systems and the large integration of intermittent renewable sources boosted the demand for efficient thermal energy storage (TES) solutions able to bridge time gaps between energy supply and demand.[1] At the same time, the increasing adoption of waste heat recovery applications, the strict thermal control requirements of power electronics, and the demand for energy-saving buildings broadened the range of applications for TES.[2]

Among the assorted options, latent heat thermal energy storage (LHTES) based on phase change materials (PCMs) has attracted particular interest.[3] This is due to its ability to absorb and release substantial amounts of thermal energy over a narrow temperature range,

© 2026 American Institute of Chemical Engineers.
All rights reserved, including rights for text and data mining and training of artificial intelligence technologies or similar technologies.

resulting in high volumetric energy density compared to conventional sensible storage systems.[4] However, the thermodynamic benefits of PCM-based LHTES systems are fundamentally constrained by the low thermal conductivity of most organic PCMs, particularly paraffin compounds, despite these thermodynamic benefits.[5] The property deficiency manifests as slow heat penetration during charging and discharging, lagging solid–liquid interface growth, and high thermal non-uniformity within the storage domain.[6] Various methodologies can be adopted to enhance thermal conductivity, such as metallic fins, nanoparticle dispersions, expanded graphite matrices, microencapsulation, and open-cell metallic foams.[7] More recently, the integration of PCMs with bio-inspired architecture and artificial intelligence-assisted design has emerged as a promising approach for optimizing heat transfer pathways, temperature uniformity, and overall thermal storage performance.[8]

Among these enhancement strategies, metal foam-PCM composites have received extensive attention because the foam's densely interconnected ligament network greatly increases the effective thermal conductivity and offers better lateral heat redistribution.[9] In conventional metal foams, the pore structures are intrinsically stochastic. This makes it difficult to tune key structural parameters such as porosity independently, mean pore diameter, specific surface area, and flow paths, thereby hindering systematic performance optimization.[10] Architected materials and metamaterials have also attracted increasing attention in energy applications because their geometrical architecture can be tailored to control energy transport and enhance functional performance. Govindaraman et al.[11] reviewed different metamaterial concepts for energy harvesting and highlighted the important role of engineered geometry in improving energy interaction and conversion. This broader design concept also motivates the use of TPMS architectures, whose controlled topology and high interfacial area provide promising opportunities for thermal energy management applications.

Triply periodic minimum surfaces (TPMS) based metallic lattices offer a mathematically rigorous and geometrically controlled alternative to stochastic foams.[12] TPMS structures are defined by implicit periodic functions and possess smooth, continuously curved, and fully interconnected surfaces.[13,14] The architectural parameters, such as porosity, wall thickness, unit-cell periodicity, and surface area to volume ratio, can be tuned individually and accurately.[15] Hence, topologies such as Gyroid, Schwarz Diamond, Schwarz Primitive, IWP, and Neovius surfaces have attracted increasing attention for applications in latent heat storage, passive thermal regulation, electronic cooling, compact heat exchanger design, and packed-bed thermal energy storage systems.[16] In fact, Chtourou et al.[17] used the Gyroid-based TPMS structure to enhance the heat transfer in compact plate heat exchangers. They compared the Gyroid configuration with a conventional chevron-type plate under the same operating conditions for a Reynolds number varying between 1000 and 2500, using three-dimensional CFD simulations in ANSYS Fluent R19. Their results showed that Gyroid TPMS geometry improves thermo-hydraulic performance with a higher Nusselt number and moderate pressure drop. The overall performance of the Gyroid-based design was improved by 16.85% compared to the conventional chevron plate. Extending the application of Gyroid architectures to latent heat storage, Torri et al.[18] experimentally and numerically investigated PCM solidification in additively manufactured Gyroid heat exchangers with porosities ranging from 70% to 80%. The results show that reducing the porosity from 80% to 70% shortened the solidification time by nearly 40%, among other ways, highlighting the key role of porosity in balancing heat-transfer enhancement, latent storage capacity, and hydraulic performance.

Previous studies investigated various configurations of TPMS-PCM composites numerically and experimentally and demonstrated their potential.[19,20] Qureshi et al.[21] observed a 31%–40% lower melting time for Gyroid, IWP, and Primitive compared to Kelvin-cell foam, which proved the fact that the thermal performance depends on the topology of the conductive skeleton and not on the porosity alone. Building upon these advances, Yan et al.[21] combined three-dimensional numerical simulations based on the enthalpy-porosity method with visual experiments to investigate the melting behavior of paraffin embedded in additively manufactured TPMS structures. Fifteen designs were evaluated by varying the topology, such as Diamond, Gyroid, Primitive, Porosity (70%–90%), and pore-gradient distribution. The Diamond structure exhibited the best thermal performance, reducing the melting time by 8.7% and 5.1% compared with the Gyroid and Primitive configurations, respectively. Moreover, a positive porosity gradient increased the average heat-transfer coefficient by 8.1%, while the experimentally tested Diamond composite shortened the melting time from 2179 s for pure paraffin to 320 s, corresponding to an 85.3% reduction. These findings highlight the importance of jointly optimizing topology, porosity, and pore distribution to enhance the thermal performance of TPMS-PCM systems. Besides, Nie et al.[22] reported that Diamond structures achieved the most favorable melting enhancement, reducing melting time by 47%–75.5% and increasing power density by 178%–275% compared with Kelvin foam. Their findings also indicate that higher porosity weakens conduction and larger pore size promotes natural convection. In addition, Gado et al.[23] showed that Gyroid heat sinks are more effective than IWP and Primitive designs for managing heat in electronic devices. However, they also pointed out that increasing the relative density by 10%–20% does not significantly improve thermal performance. Hence, the Neovius topology showed the fastest transient response and highest effective thermal conductivity in microgravity environments, where convection is suppressed.[24] Moreover, Al-Safadi et al.[25] performed a CFD comparison of the heat transfer between TPMS lattices, non-TPMS lattices, and fins. They observed that the fastest melting was obtained with traditional fins with isothermal heating, while the performance of IWP and Primitive structures was competitive, especially under iso-flux conditions. Thus, the results reported that phase change dynamics depend on lattice topology, heating temperature, and orientation, where Gyroid and IWP have better melting and solidification behavior.[25]

The effect of inclination on TPMS-PCM systems was studied by Yang et al.,[26] who found that a moderate tilt can enhance natural convection. However, it may also degrade conduction, and thus the

19447450, 0, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/ep.70734 by Mahidol University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

system is sensitive to the operating position. Furthermore, Huang et al.[27] proposed a field-driven, spatially graded porosity design strategy for TPMS structures, which resulted in phase change time reductions relative to uniform porosity configurations and an increase in total heat storage capacity. Also, Zhang et al.[28] demonstrated TPMS-based PCM capsules for electronic cooling, which showed over 70% reduction in melting time and up to 97.2% increase in peak thermal charging power compared to conventional spherical capsules, but at the cost of significantly higher pressure drop. Extending this optimization framework, Huang et al.[27] combined numerical simulations with visualization experiments to investigate the melting performance of PCM composites incorporating Gyroid, Diamond, and Primitive TPMS structures with porosities ranging from 60% to 80%. A temperature field-driven design strategy was implemented using various topologies to generate nonlinear porosity distributions that adapt the conductive skeleton to local thermal conditions. Compared with the corresponding uniform structures, the optimized configurations shortened the complete melting time by 68.4, 81, and 44 s for the Gyroid, Diamond, and Primitive topologies, respectively, while increasing their heat-storage capacities by 64, 37, and 96 J. These results demonstrate that field-driven porosity redistribution can effectively balance melting enhancement and energy-storage capacity in TPMS-based latent heat storage systems.

Although several studies have investigated TPMS-based PCM systems, the combined influence of topology and porosity on the balance between heat-transfer enhancement and latent heat storage remains insufficiently quantified under consistent operating conditions. The present study therefore provides a systematic comparative assessment of TPMS-based structures embedded in RT31 PCM under identical thermal boundary conditions. Twelve configurations based on Gyroid, Schwarz Diamond, and Schwarz Primitive topologies are examined. To isolate the influence of topology, the three geometries are directly compared at a common porosity of 85.11%, while the effect of porosity is subsequently evaluated separately within each TPMS family. The assessment combines melting time, liquid-fraction evolution, thermal uniformity, buoyancy-driven flow development, and latent heat storage capacity. This multi-criteria approach enables the identification of a balanced configuration rather than considering melting rate alone. The main objective is therefore to clarify how TPMS topology and porosity jointly influence the competing mechanisms of heat transfer, natural convection, and energy-storage capacity in RT31-based latent heat thermal storage systems.

## 2 | MATERIALS AND METHODS

### 2.1 | Physical assumptions

To simplify the numerical model while maintaining the PCM's melting behavior in a physically realistic manner, several assumptions were made. These assumptions simplify the governing equations of the system and are listed below:

- Transient Regime: Since one of the objectives of this study is to characterize the melting behavior of PCM, the temporal evolution of the main variables must be considered
- Laminar Regime: Since the considered volumes and geometries are relatively small, the velocities that develop inside the PCM remain limited. The heat transfer and fluid flow are therefore assumed to stay in the laminar regime. The laminar-flow assumption was further assessed using the Rayleigh number:

$$Ra = \frac{g\beta\Delta T L^3}{\nu\alpha} \tag{1}$$

where $g$ is gravitational acceleration ($m.s^{-2}$), $\beta$ is the thermal expansion coefficient ($K^{-1}$), $\Delta T$ is the characteristic temperature difference *(K)*, $L$ is the characteristic length (m), $\nu$ is the kinematic viscosity ($m^2.s^{-1}$), and $\alpha$ is the thermal diffusivity ($m^2.s^{-1}$).

$$\nu = \frac{\mu}{\rho} \tag{2}$$

$$\alpha = \frac{k}{\rho C_p} \tag{3}$$

Using the RT31 properties, a characteristic length of $L = 3$ mm, and $\Delta T = 11$ K corresponding to the considered melting temperature interval relative to the initial condition, the characteristic Rayleigh number is approximately $1.72 \times 10^3$. Since the same PCM properties, characteristic length, and thermal boundary conditions are used for all twelve configurations, this nominal Rayleigh number is identical for all cases. The relatively low Rayleigh number supports the assumption of laminar natural convection.

- Boussinesq approximation: This approximation assumes that density variations are small enough to be neglected everywhere except in the buoyancy term, where they are responsible for natural convection
- Incompressible flow: Because of the Boussinesq approximation, density variations are neglected in the continuity equation, and the flow is therefore considered incompressible.
- Thermophysical properties: The thermophysical properties of both aluminum and PCM were assumed to be constant, except for the PCM density, whose temperature dependence was modeled using the Boussinesq approximation given in Equation (6).
- No viscous dissipation: the heat generated by viscous dissipation in the PCM is considered negligible compared with the dominant heat transfer mechanisms, namely convection and conduction.
- No radiation heat transfer: Radiative heat transfer is considered negligible and is therefore not considered in the thermal analysis.
- No-slip condition at solid walls: The velocity of the PCM is assumed to be zero at all solid boundaries
- Enthalpy-porosity formulation: The phase change process is modeled using the enthalpy-porosity method. In this approach, the liquid fraction is calculated from the local temperature, and the

19447450, 0, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/ep.70734 by Mahidol University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

mushy zone is treated as a porous medium. A momentum sink term is introduced in partially or completely solid regions to suppress the velocity as the PCM solidifies progressively.

- The mushy-zone constant was set to $A_{mush} = 10^5$, which is commonly adopted in the enthalpy-porosity method and has provided good agreement with experimental PCM melting data in previous studies.[28–30]
- Perfect thermal contact was assumed between the TPMS surfaces and the PCM. Therefore, thermal contact resistance and possible interfacial air gaps were neglected. PCM volume change during melting was also neglected under the incompressible Boussinesq formulation. In contrast, the restriction of buoyancy-driven flow in narrow TPMS passages is directly represented by the resolved geometry and the no-slip boundary condition.

## 2.2 | Governing equations

Under the previously stated assumptions, the governing equations of the present physical problem are presented in this section. For clarity, the general conservation equations are first presented, followed by the simplified forms that reflect the assumptions adopted in this study.

*Mass Conservation Equation*: The general mass conservation equation is written as:

$$\frac{\partial \rho}{\partial t} + \nabla.\left(\rho \vec{V}\right) = 0 \tag{4}$$

where $\rho$ is the density (kg.m$^{-3}$) and $\vec{V}$ is the velocity vector (m.s$^{-1}$).

*Momentum Conservation Equation*: The general momentum conservation equation for a Newtonian fluid is given by:

$$\frac{\partial\left(\rho \vec{V}\right)}{\partial t} + \nabla.\left(\rho \vec{V}\vec{V}\right) = -\nabla p + \nabla\left[\mu\left(\nabla \vec{V} + \left(\nabla \vec{V}\right)\right)^T\right] + \rho \vec{g} \tag{5}$$

where $p$ is the pressure (Pa), $\mu$ is the dynamic viscosity (Pa.s), and $\vec{g}$ is the gravitational acceleration vector (m.s$^{-2}$).

In addition, the Boussinesq approximation allows density variations to be neglected in all terms except the buoyancy term, where it is expressed as:

$$\rho(T) \approx \rho_0[1 - \beta(T - T_{ref})] \tag{6}$$

where $\rho_0$ is the reference density (kg.m$^{-3}$), $\beta$ is the thermal expansion coefficient (K$^{-1}$), and $T_{ref}$ is the reference temperature (K). The enthalpy-porosity formulation introduces a momentum sink term $\vec{S}$ in the solid and mushy regions, progressively suppressing the velocity. The momentum conservation equation therefore becomes:

$$\rho_0\left(\frac{\partial \vec{V}}{\partial t} + \vec{V}.\nabla\vec{V}\right) = -\nabla p + \mu\nabla^2\vec{V} + \rho(T)\vec{g} + \vec{S} \tag{7}$$

where $\vec{S}$ is the momentum sink term (N.m$^{-3}$) used to suppress velocity in the solid and mushy regions. The phase change process is modeled using the enthalpy-porosity method. In this approach, the liquid fraction $\lambda$ is determined from the local temperature field as follows:

$$\lambda = 0 \rightarrow T < T_s : \text{solid PCM.}$$

$$\lambda = \frac{T - T_s}{T_l - T_s} \rightarrow 0 < \lambda < 1 \rightarrow T_s \le T \le T_l : \text{mushy zone.}$$

$$\lambda = 1 \rightarrow T > T_l : \text{liquid PCM.}$$

where $T_s$ is the solidus temperature (K), and $T_l$ is the liquidus temperature (K).

To account for the progressive suppression of velocity in the solid and mushy regions, a momentum sink term is introduced into the momentum equation:

$$\vec{S} = -C\frac{(1-\lambda)}{\lambda^3 + \varepsilon} \tag{8}$$

where $C$ is the mushy zone constant, and $\varepsilon$ is a small number introduced to avoid division by zero. The mushy-zone constant was set to $A_{mush} = 10^5$.[31] The liquid fraction was calculated using a piecewise-linear interpolation across the phase-change interval $300.15\,K \le T \le 304.15\,K$. No additional liquid-fraction smoothing function was applied. A separate sensitivity analysis of $A_{mush}$ was not conducted in the present study.

*Energy conservation equation*: The general energy conservation equation is written as:

$$\frac{\partial(\rho H)}{\partial t} + \nabla.\left(\rho \vec{V} H\right) = \nabla.(k\nabla T) + S_h \tag{9}$$

where $H$ is the total enthalpy, $k$ is the thermal conductivity, and $S_h$ is a possible volumetric heat source term. In the present study, no additional volumetric heat source is considered, so the source term $S_h$ is neglected.

## 2.3 | Computational domain

### 2.3.1 | Triply periodic minimal surface geometries

Triply Periodic Minimal Surfaces (TPMS) are geometrically complex porous structures generated by periodically repeating a unit cell in the three spatial directions. They are typically defined numerically using implicit mathematical equations that describe a continuous surface in space. They are characterized by zero mean curvature, the mean curvature A at any point of the surface:

$$A = \frac{k_1 + k_2}{2} = 0 \tag{10}$$

19447450, 0, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/ep.70734 by Mahidol University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

In the present study, the unit-cell size was kept constant so that the effects of topology and porosity could be examined independently. At a given porosity, reducing the unit-cell size increases the specific interfacial area and shortens characteristic heat-transfer distances, which may enhance conduction-dominated melting. Conversely, larger unit cells provide wider PCM flow passages and may promote buoyancy-driven convection. Therefore, the reported topology-porosity trends are specific to the selected unit-cell size.

The properties of TPMS structures include smooth and continuous surfaces and a high surface-area-to-volume ratio. They are also characterized by high interconnectivity, continuous conductive pathways, and interconnected pores, which are especially useful for enhancing heat transfer in PCM-based systems. In addition, TPMS structures allow controlled porosity, which makes it possible to adjust the balance between heat conduction through the solid matrix and PCM storage inside the porous volume.

Among the different TPMS geometries, the Schwarz Primitive, Schwarz Diamond, and Gyroid surfaces are among the most investigated structures in thermal and energy-related applications. These geometries are typically defined using implicit level-set equations, in which the constant $c$ can be adjusted to control key structural parameters such as relative density, wall thickness, and porosity.[32] The unit-cell morphologies of the three selected TPMS geometries are shown in Figure 1.

- *Schwarz Primitive (P)*: It is characterized by a highly connected periodic geometry with a rounded, almost spherical pore structure. Its topology creates a large heat exchange surface, which makes it relevant for applications involving heat transfer and fluid flow. The implicit equation of the Schwarz P surface can be written as:

$$f(x, y, z) = \cos(2\pi x) + \cos(2\pi y) + \cos(2\pi z) = c \tag{11}$$

- *Schwarz Diamond (D)*: It presents a diamond-like interconnected topology. This structure is known for its balanced connectivity and mechanical robustness. It can be described using the following implicit equation:

$$\begin{aligned} f(x, y, z) = &\sin(2\pi x)\sin(2\pi y)\sin(2\pi z) \\ &+ \sin(2\pi x)\cos(2\pi y)\cos(2\pi z) \\ &+ \cos(2\pi x)\sin(2\pi y)\cos(2\pi z) \\ &+ \cos(2\pi x)\cos(2\pi y)\sin(2\pi z) = c \end{aligned} \tag{12}$$

- *Gyroid (G)*: It is characterized by a continuous and spiral-like topology. Its interconnected channels make it particularly attractive for thermo-fluid applications, since it can promote fluid transport while maintaining a large heat exchange surface. The Gyroid surface can be expressed as:

$$f(x, y, z) = \sin(2\pi x)\cos(2\pi y) + \sin(2\pi y)\cos(2\pi z) + \sin(2\pi z)\cos(2\pi x) = c \tag{13}$$

In these equations, x, y, and z are normalized spatial coordinates.

The TPMS geometries were generated from the implicit equations of the Gyroid, Schwarz Diamond, and Schwarz Primitive surfaces. The level-set parameter $c$ was varied to obtain the required porosity for each configuration. The generated unit cells were scaled to the computational-domain dimensions and used to define the internal TPMS-PCM interfaces. The final geometry was then imported into ANSYS Fluent for numerical simulation.

### 2.3.2 | Geometrical parameters of TPMS

- *Porosity* $\varepsilon$: Porosity represents the fraction of void volume in a TPMS structure. The higher the porosity, the smaller the volume occupied by the TPMS solid matrix in the system. It is defined as:

$$\varepsilon = \frac{V_{void}}{V_{total}} = 1 - \frac{V_{solid}}{V_{total}} \tag{14}$$

where $V_{void}$ is the volume of the void space available for the fluid ($mm^3$), $V_{solid}$ is the volume of the TPMS ($mm^3$), and $V_{total}$ is the total volume of the structure ($mm^3$). Porosity controls the balance between the amount of PCM available for latent heat storage and the amount of solid conductive material available for heat diffusion.

- *Wall thickness t (m)*: Wall thickness represents the thickness of the solid TPMS structure. Increasing the wall thickness reduces porosity, which can improve heat conduction but also reduces the volume available for PCM storage.
- *Unit cell size L (m)*: Unit cell size represents the characteristic dimension of the elementary TPMS cell repeated in space. For a given

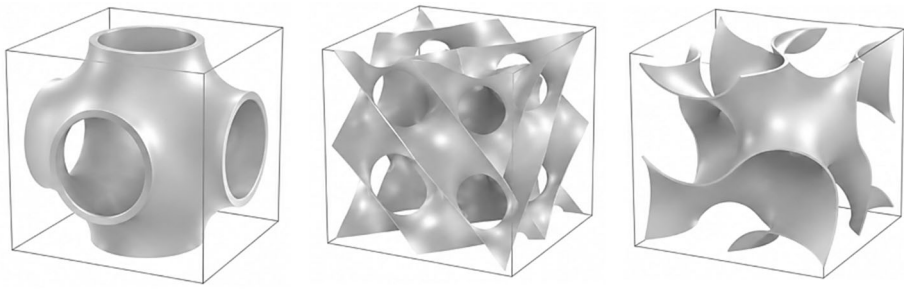



**FIGURE 1** TPMS unit cells: (a) Schwarz primitive, (b) Schwarz diamond, and (c) Gyroid.

19447450, 0, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/ep.70734 by Mahidol University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

porosity, changing the unit cell size can modify the specific surface area, therefore influencing the heat transfer behavior of the structure.

- *Tortuosity* $\tau$: Tortuosity describes the complexity of the flow path inside the porous structure. It is related to the effective path length for heat transfer and fluid flow, which can reduce effective thermal conductivity but also enhance convective mixing.

### 2.3.3 | Studied configurations

In the present model, the skeleton of the TPMS is represented by heated internal surfaces contained in the PCM domain with dimensions of $3 \times 3 \times 3\ mm^3$. This method considers the dominant heat transfer interface between the conductive aluminum structure and the PCM and reduces the computational cost.

In the adopted approach, the aluminum TPMS is not modeled as a volumetric solid domain. Therefore, the approach does not explicitly resolve transient heat storage and spatial temperature gradients within the aluminum skeleton. Instead, heat is supplied directly through the TPMS-PCM interface using a prescribed uniform heat flux. This approximation is justified by the high thermal conductivity of aluminum relative to RT31 PCM and the small characteristic dimensions of the TPMS structure. Nevertheless, this approach represents an idealized heating condition and is primarily intended for the comparative assessment of topology and porosity under identical boundary conditions. A fully conjugate solid-PCM model was required to resolve the detailed temperature distribution and conduction resistance within the aluminum skeleton.

A geometrical characterization of each configuration was performed to help the interpretation of the thermal behavior of the TPMS-PCM systems. The porosity $\varepsilon$ (%), aluminum volume fraction, PCM volume fraction, PCM volume, and aluminum volume were considered as primary indicators. Table 1 presents the different geometrical characteristics of the investigated twelve configurations. These indicators enable the relationship between melting behavior and the balance between conductive heat transfer in the aluminum skeleton and latent heat storage capacity in the PCM volume.

The specific surface area (SSA) is defined as the TPMS-PCM interfacial area normalized by the total computational volume:

$$SSA = \frac{A_{TPMS}}{V_{total}} \tag{15}$$

where $A_{TPMS}$ is the TPMS-PCM interfacial area and $V_{total}$ is the total domain volume. In the present numerical approach, the TPMS geometry is represented by zero-thickness internal surfaces; consequently, a finite wall thickness is not explicitly modeled.

This parameter provides a quantitative measure of the TPMS-PCM interfacial area available for heat transfer per unit PCM volume. As the porosity decreases, the TPMS surface area generally increases while the PCM volume decreases, resulting in a higher SSA. However, SSA alone does not fully govern the melting performance, as the TPMS topology also influences three-dimensional channel connectivity, characteristic pore dimensions, heat-transfer pathways, and buoyancy-driven flow.

The studied configurations G1–G4, D1–D4, and P1–P4 denote increasing/decreasing porosity cases within each topology. Consequently, direct topology comparison is performed only for G3, D3, and P3 at the identical porosity of 85.11%, whereas the remaining configurations are used exclusively to assess porosity effects within each TPMS family.

The numerical campaign consists of 12 TPMS configurations, obtained by combining three unit-cell topologies: Gyroid, Schwarz D, and Schwarz P, with four porosity levels for each geometry, as described in Figure 2.

## 2.4 | Boundary conditions

The present approach is not a fully conjugate solid–fluid model. Heat transfer, natural convection, and phase change are solved within the PCM domain, whereas the aluminum TPMS is represented by heated

**TABLE 1** Geometrical characteristics of the investigated TPMS-PCM configurations.

| Config | Porosity $\varepsilon$ (%) | PCM volume fraction (%) | Aluminum volume fraction (%) | PCM volume $V_{PCM}$ ($mm^3$) | Aluminum volume $V_{Al}$ ($mm^3$) | SSA ($m^{-1}$) |
|---|---|---|---|---|---|---|
| G1 | 92.03 | 92.03 | 7.97 | 24.848 | 2.152 | ~1071 |
| G2 | 87.33 | 87.33 | 12.67 | 23.579 | 3.421 | ~1187 |
| G3 | 85.11 | 85.11 | 14.89 | 22.980 | 4.020 | ~1254 |
| G4 | 78.18 | 78.18 | 21.82 | 21.109 | 5.891 | ~1522 |
| D1 | 90.07 | 90.07 | 9.93 | 24.319 | 2.681 | ~1193 |
| D2 | 87.74 | 87.74 | 12.26 | 23.690 | 3.310 | ~1266 |
| D3 | 85.11 | 85.11 | 14.89 | 22.980 | 4.020 | ~1371 |
| D4 | 76.30 | 76.30 | 23.70 | 20.601 | 6.399 | ~1699 |
| P1 | 92.78 | 92.78 | 7.22 | 25.051 | 1.949 | ~998 |
| P2 | 89.11 | 89.11 | 10.89 | 24.060 | 2.940 | ~1101 |
| P3 | 85.11 | 85.11 | 14.89 | 22.980 | 4.020 | ~1218 |
| P4 | 83.44 | 83.44 | 16.56 | 22.529 | 4.471 | ~1287 |

19447450, 0, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/ep.70734 by Mahidol University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

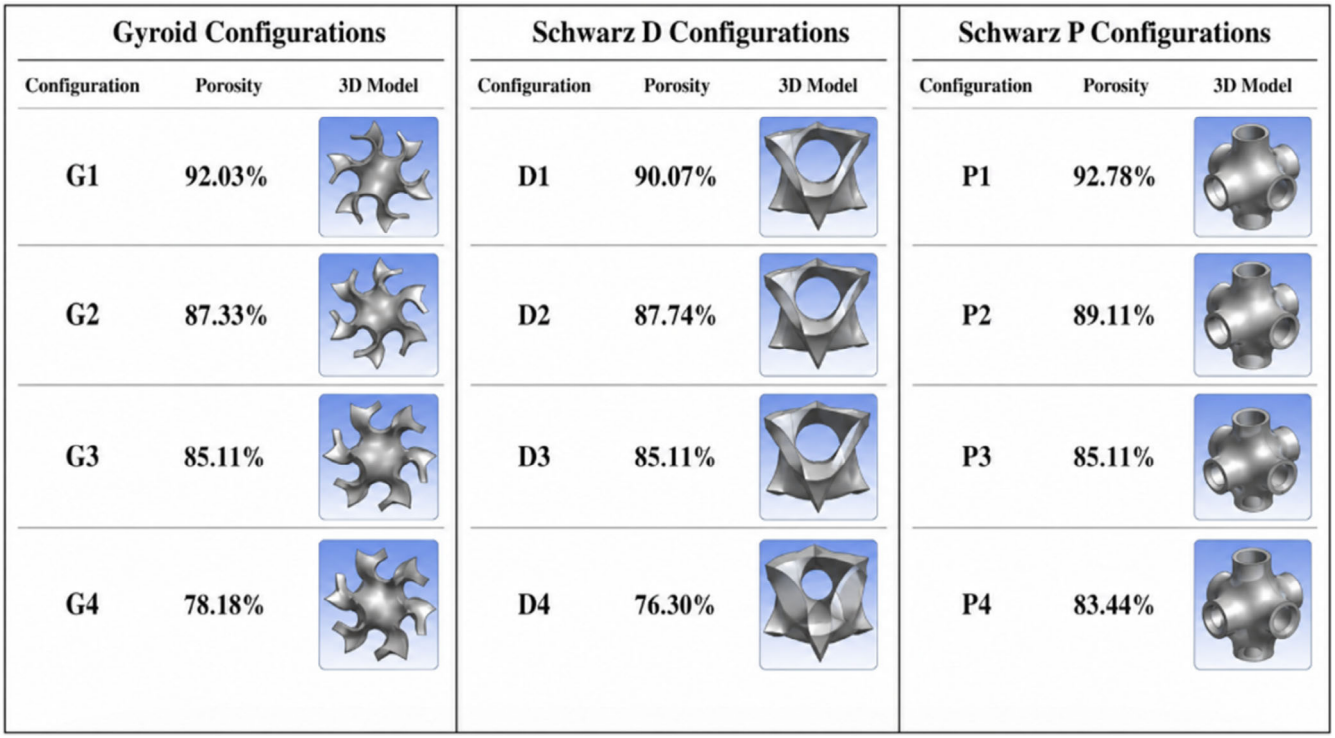



**FIGURE 2** Investigated Gyroid, Schwarz diamond, and Schwarz primitive configurations with their corresponding porosity levels.

**TABLE 2** Thermophysical properties of RT31 PCM used in ANSYS Fluent.[29]

| Property | Value | Unit |
|---|---|---|
| Density $\rho$ | 830 | $Kg.m^{-3}$ |
| Specific heat $C_p$ | 2000 | $J.kg^{-1}.K^{-1}$ |
| Thermal conductivity $k$ | 0.2 | $W.m^{-1}.K^{-1}$ |
| Dynamic viscosity $\mu$ | 0.03499 | $Kg.m^{-1}.s^{-1}$ |
| Thermal expansion coefficient $\beta$ | 0.003 | $K^{-1}$ |
| Latent heat of fusion $L$ | 165,000 | $J.kg^{-1}$ |
| Solidus temperature $T_s$ | 302.15 | K |
| Liquidus temperature $T_l$ | 304.15 | K |

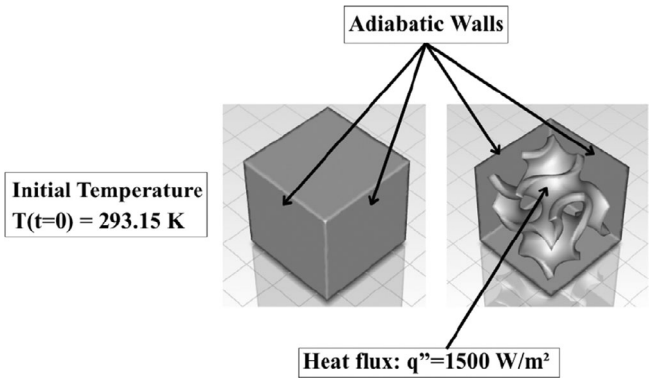



**FIGURE 3** Boundary and initial conditions applied to the computational domain: Gyroid example.

internal surfaces. Consequently, conduction and thermal inertia within a volumetric aluminum skeleton are not explicitly modeled.

The single-domain approach allows for imposing the thermal boundary conditions on the faces of the TPMS directly without the need for a fully coupled solid–fluid domain, thereby decreasing the computational cost while maintaining the main heat-transfer features of the PCM-TPMS system.

The PCM used in all simulations was RT31, a commercial paraffin wax with a peak melting temperature of 31°C. Most of its thermophysical properties were obtained from the manufacturer's material data sheet.[29] Table 2 shows the values used for simulations.

RT31 undergoes a phase change over the temperature range 302.15 – 304.15K, corresponding to the solidus and liquidus temperatures, respectively. In the present model, the thermophysical properties were assumed constant. Density variations with temperature were considered only in the buoyancy term through the Boussinesq approximation.

The TPMS skeleton was represented by zero-thickness internal surfaces within the PCM domain. A uniform heat flux of $1500 W m^{-2}$ was imposed directly on these surfaces. This boundary condition represents an idealized heat supply through the aluminum TPMS structure while maintaining identical thermal conditions for all investigated configurations.

Figure 3 summarizes the boundary and initial conditions imposed on the computational domain.

19447450, 0, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/ep.70734 by Mahidol University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

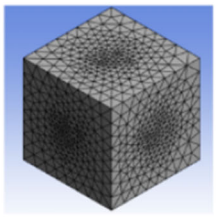
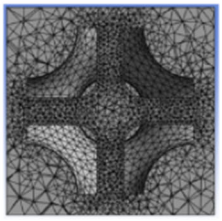
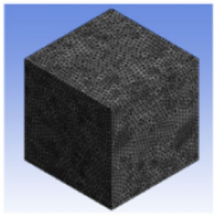
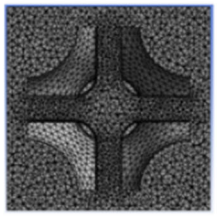
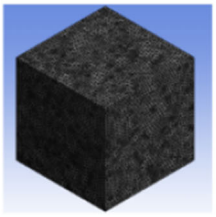
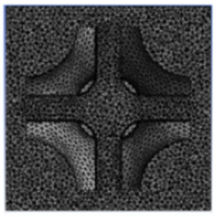
| Mesh | Mesh size (mm) | Mesh Element number | Meshed domain description | |
|---|---|---|---|---|
| | | | 3D view | Cross-section view |
| Coarse mesh | 0.5 | 63553 |  |  |
| Medium mesh | 0.1 | 215627 |  |  |
| Fine mesh | 0.08 | 392083 |  |  |

**FIGURE 4** Mesh sensitivity study for the P4 configuration.

### 2.4.1 | TPMS internal facets

The boundary conditions associated with the TPMS structures are imposed on the internal faces of the PCM volume resulting from geometric subtraction.

A uniform local heat flux of $q'' = 1500\,\mathrm{W\,m^{-2}}$ was imposed on the TPMS–PCM interfaces for all configurations. Since the TPMS interfacial area varies with topology and porosity, the resulting total thermal power is not strictly identical among the investigated cases. Therefore, the results should be interpreted as a comparison under constant heat-flux-density conditions rather than constant total heat input.[33]

*External walls*

To isolate the effect of the TPMS on the thermal behavior of the PCM, the only heat source considered in the model is the TPMS internal faces. Therefore, the external walls are assumed to be perfectly adiabatic. This condition can be expressed as: $\frac{\partial T}{\partial n}\big|_{\partial\Omega_{ext}} = 0$ meaning that no heat flux crosses $\partial\Omega$, the external boundary of the PCM domain.

*Initial conditions*

At $t = 0$, the domain is considered at ambient temperature: $T\,(t = 0) = 293.15$ K. Since the melting point of the material considered, RT31, is above this initial temperature, the PCM is initially fully solid. The initial liquid fraction is therefore: $\lambda\,(t = 0) = 0$.

## 2.5 | Mesh generation

Since the PCM-TPMS configurations involve complex curved internal surfaces, an unstructured tetrahedral mesh was generated for the computational domain. Tetrahedral elements are well suited to such geometries because they can conform efficiently to highly curved and irregular interfaces. Local mesh refinement was applied near the TPMS-PCM interfaces to better resolve the steep thermal gradients developing during the melting process.[34,35]

The P4 configuration was selected as a representative case for the mesh-independence assessment. The mesh size identified from this analysis was subsequently applied consistently to all TPMS configurations using the same meshing strategy and quality criteria. Therefore, the mesh study was intended to establish an adequate global spatial resolution rather than to perform a topology-specific mesh optimization.

A mesh sensitivity study was carried out using one representative geometry: the P4 configuration, corresponding to an 83.44% Schwarz P geometry. Three mesh sizes were considered: a coarse mesh, a medium mesh, and a fine mesh, as described in Figure 4. To capture the influence of mesh size on the simulation result, the volume-averaged PCM temperature was compared for the three meshes.

The mesh sensitivity results indicate that the medium (0.10 mm, 215,627 elements) and fine (0.08 mm, 392,083 elements) meshes give very similar temperature evolutions, especially after 100 s, where the two curves nearly coincide and reach a final mean PCM temperature of about 356 K (Figure 5). Therefore, the

19447450, 0, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/ep.70734 by Mahidol University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

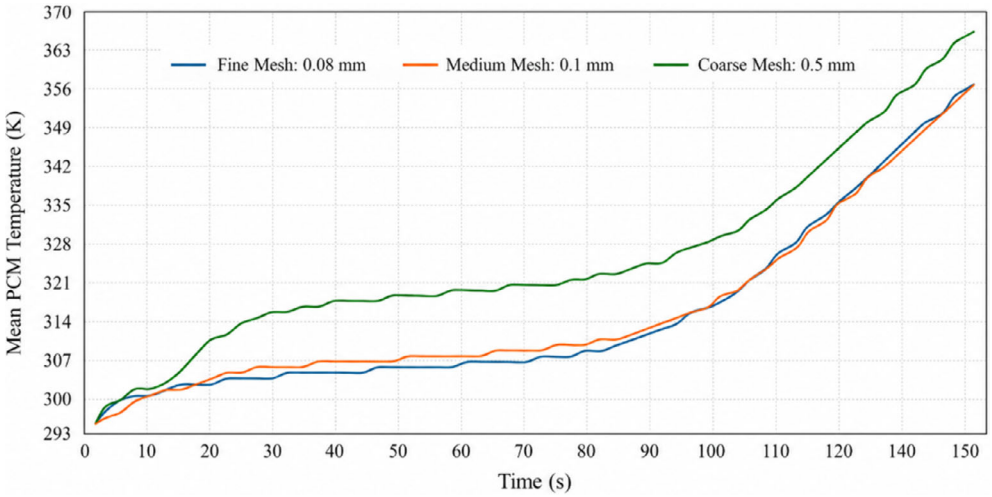



**FIGURE 5** Temporal evolution of the mean PCM temperature for the three different mesh sizes at time step $\Delta t = 0.5$ s.

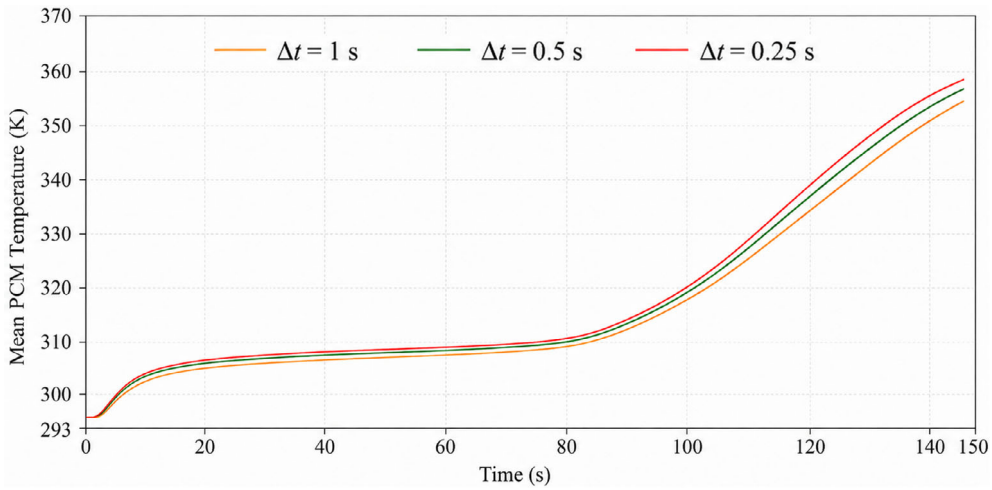



**FIGURE 6** Temporal evolution of the mean PCM temperature for the three different time steps $\Delta t = 1$, 0.5, and 0.25 s.

medium mesh was selected as a good compromise between numerical accuracy and computational cost.

## 2.6 | Solver fluent parameters

The numerical simulations were conducted in ANSYS Fluent R19 utilizing a pressure-based transient solver. The energy equation, gravitational forces, and the solidification-melting model were employed to address heat transfer, buoyancy-driven flow, and phase transitions within the PCM domain. The spatial gradients were computed employing the least-squares cell-based method, while the pressure, momentum, and energy equations were discretized utilizing second-order schemes to enhance numerical precision.[4]

The time-step sensitivity study indicates that the three different time steps tested, $\Delta t = 1$ s, 0.5 s, and 0.25 s, yield very similar average PCM temperature evolutions during the early and intermediate melting stages, as illustrated in Figure 6. The difference between $\Delta t = 0.5$ s and $\Delta t = 0.25$ s is overall limited, which demonstrates that further time-step refinement does not significantly change the predicted thermal behavior. Consequently, $\Delta t = 0.5$ s was selected as a suitable compromise between numerical accuracy, temporal stability, and computational cost.

## 2.7 | Validation model

Before analyzing the simulation results, it is important to first validate the numerical model to ensure the reliability of the obtained results. The validation was performed against experimental data for RT31 PCM using $N = 13$ paired temperature data points from Arqam et al.[36] The error indicators are defined as:

$$MAE = \frac{|T_{num} - T_{exp}|}{T_{exp}} \times 100 \quad (16)$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(T_{num,i} - T_{exp,i}\right)^2} \quad (17)$$

19447450, 0, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/ep.70734 by Mahidol University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

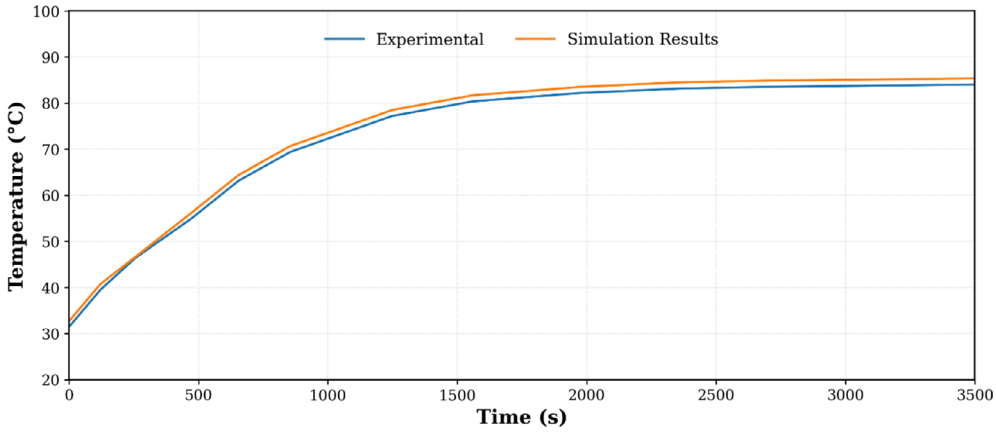



**FIGURE 7** Comparison between the present numerical and the previous experimental approaches (Arqam et al.[36]) in terms of the temporal evolution of the average base temperature.

**TABLE 3** Statistical indicators for the benchmark verification of the numerical model.

| Error indicator | Value | Unit |
|---|---|---|
| Mean absolute error (MAE) | 1.2085 | °C |
| Root mean square error (RMSE) | 1.2418 | °C |
| Coefficient of determination ($R^2$) | 0.9953 | – |
| Maximum absolute error (MaxAE) | 1.34 | °C |

$$R^2 = 1 - \frac{\sum_{i=1}^{n} \left(T_{exp,i} - T_{num,i}\right)^2}{\sum_{i=1}^{n} \left(T_{exp,i} - \overline{T}_{exp}\right)^2} \tag{18}$$

$$\text{MaxAE} = \max_{1<i<N} \left| T_{sim,i} - T_{exp,i} \right| \tag{19}$$

Figure 7 presents a comparison of the temporal base temperature evolution of RT31 PCM between the present numerical simulation and the experimental reference data. A good agreement is observed between the two curves throughout the heating process, with only slight deviations at higher temperatures. The quantitative error indicators reported in Table 3 further confirm the accuracy of the numerical predictions. The close agreement demonstrates that the adopted enthalpy–porosity model can reliably reproduce the transient thermal behavior of RT31 PCM.

The model yielded an MAE of 1.21°C, an RMSE of 1.24°C, a maximum absolute error of 1.34°C, and $R^2 = 0.995$, confirming good agreement with the experimental RT31 data. Although the experimental configuration differs from the present TPMS geometry and boundary conditions, this validation primarily assesses the ability of the enthalpy–porosity model to reproduce the transient thermal behavior of RT31 PCM.

## 3 | RESULTS

### 3.1 | Effect of TPMS topology on PCM melting

In this section, the effect of the TPMS topology on the melting behavior of the PCM was presented by comparing three configurations with the same porosity level: Gyroid (G3), Schwarz Diamond (D3), and Schwarz Primitive (P3). The aim is to isolate the effect of the internal geometrical architecture on heat-transfer enhancement, liquid fraction evolution, the temperature distribution, and melting propagation. This comparison allows revealing the most effective TPMS structure for speeding up the PCM melting under the same thermal conditions. The studied configurations (G3, D3, and P3) were at the same porosity level of 85.11%, as shown in Figure 8.

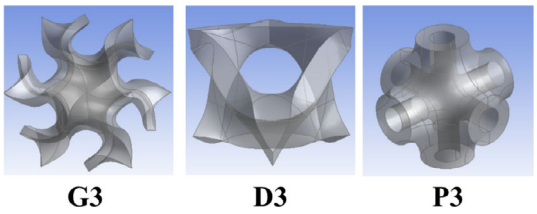



**FIGURE 8** Selected configurations for geometry comparison; G3: Gyroid, D3: Schwarz D, and P3: Schwarz P at porosity level of 85.11%.

One of the main objectives of integrating TPMS-based metal foams into the PCM system is to accelerate the melting process. Therefore, it is important to compare the melting time associated with each geometry to identify the most effective one for achieving this goal. A first global indication for the comparison of the TPMS geometries is the temporal evolution of the volume-averaged PCM liquid fraction.

To quantitatively assess thermal uniformity, the temperature spread and normalized temperature non-uniformity were defined as:

$$\Delta T = T_{\max} - T_{\min} \tag{20}$$

$$U_T = \frac{T_{\max} - T_{\min}}{T_{\text{avg}}} \times 100 \tag{21}$$

where $T_{\max}$, $T_{\min}$, and $T_{\text{avg}}$ denote the maximum, minimum, and volume-averaged PCM temperatures, respectively. Lower values of $\Delta T$ and $U_T$ indicate better thermal uniformity.

In this study, the characteristic melting time $t_{95}$ is defined as the time required for the volume-averaged PCM liquid fraction to reach

19447450, 0, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/ep.70734 by Mahidol University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

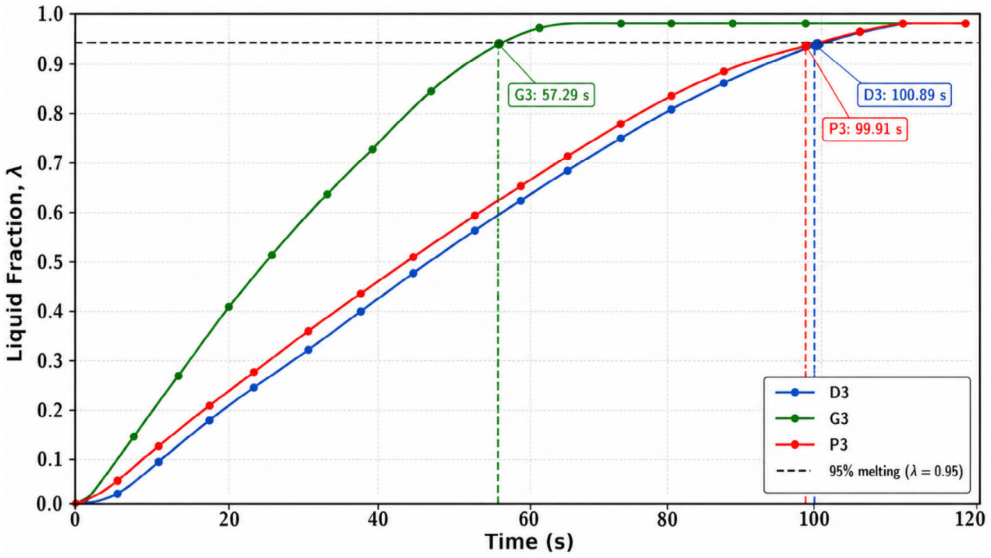



**FIGURE 9** Temporal evolution of the volume-averaged liquid fraction for configurations G3, D3, and P3 at the same porosity level of 85.11%.

$\lambda = 0.95$. This near-complete melting criterion avoids sensitivity associated with small residual solid regions during the final stage of melting and provides a consistent basis for comparison between configurations.

Figure 9 presents the evolution of the PCM liquid fraction over time for the three TPMS models D3, G3, and P3 at the same porosity level of 85.11%. The Gyroid configuration G3 exhibits the most rapid melting response, attaining 95% liquid fraction at 57.29 s, whereas Schwarz D (D3) and Schwarz P (P3) achieve 95% liquid fraction at 100.89 and 99.91 s, respectively. This indicates that the Gyroid structure decreases the melting time by about 43.2% compared to Schwarz D and 42.7% compared to Schwarz P. The near-overlapping behavior of D3 and P3 indicates that both Schwarz-based geometries offer similar thermal pathways at the current operating conditions. P3 achieves $\lambda = 0.95$ at a time slightly before D3, but the difference is only about 0.98 s, or less than 1% relative deviation. This very small variation means that Schwarz D and Schwarz P have almost the same global melting performance for a porosity of 85.11%, while the Gyroid geometry is still clearly better.

The benefit of the Gyroid topology is the continuous, smooth, and highly connected channels that enhance the heat diffusion from the metallic structure to the PCM, offering good paths for the buoyancy-driven natural convection. The difference is particularly pronounced at intermediate times: at $t = 50$ s, G3 already has a liquid fraction of about 0.85–0.90, while D3 and P3 are still close to 0.50, which points to a much faster thermal penetration and phase-change propagation. However, D3 and P3 present very similar melting behavior with only a small difference of about 0.98 s at $\lambda = 0.95$, which means that their thermal performance is almost equal in the present conditions.

Overall, these findings confirm that the melting dynamics are strongly governed by the TPMS topology for the same porosity. Gyroid geometry provides a favorable combination of distributed interfacial heating and buoyancy-driven convection in the molten PCM and thus is the most efficient configuration for accelerating the process of charging of PCM-based latent heat thermal storage devices.

Figure 10 depicts the liquid-fraction contours on the XZ mid-plane for the three TPMS designs G3, D3, and P3 at the same porosity level of 85.11% during the melting phase. The findings clearly demonstrate the quicker melting behavior of the Gyroid arrangement (G3). The melted PCM region develops more rapidly around the Gyroid structure in the early stages, particularly between 10 and 30 s, showing more efficient heat diffusion from the TPMS metallic structure to the PCM. By 50–60 s, most of the PCM domain in G3 has already melted, corresponding with the quantitative result indicating that G3 reaches $\lambda = 0.95$ at around 57.29 s.

By contrast, the Schwarz D3 and Schwarz P3 layouts show slower melting-front propagation. Large patches of partially solid PCM can be seen at intermediate times, especially near the exterior corners and low-temperature zones. Even after 60–80 s, D3 and P3 still exhibit non-uniform melting patterns, although G3 has nearly reached a totally liquid state. This demonstrates that the Schwarz-based structures slow the propagation of the phase-change front compared to the Gyroid topology.

The variation between the three configurations is primarily due to the internal architecture of the TPMS structure. The Gyroid shape creates smooth, continuous, and interconnected channels that optimize the coupling of conductive heat transmission between the metallic structure and buoyancy-driven convection in the liquid PCM. This produces a more homogeneous temperature field while reducing stagnant zones. In contrast, the Schwarz D and Schwarz P structures produce more concentrated melting patches, indicating a less uniform distribution of heat and less convective circulation throughout the transient melting process.

In summary, these contour results confirm that, with the same porosity level of 85.11%, the Gyroid topology is the most efficient arrangement for speeding PCM melting. The illustration visually validates the quantitative decrease in melting time seen previously. G3 minimizes the time necessary to obtain $\lambda = 0.95$ by more than 42% compared to D3 and P3.

Although absolute melting times depend on domain dimensions and thermal loading, all cases were evaluated under identical conditions.

19447450, 0, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/ep.70734 by Mahidol University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

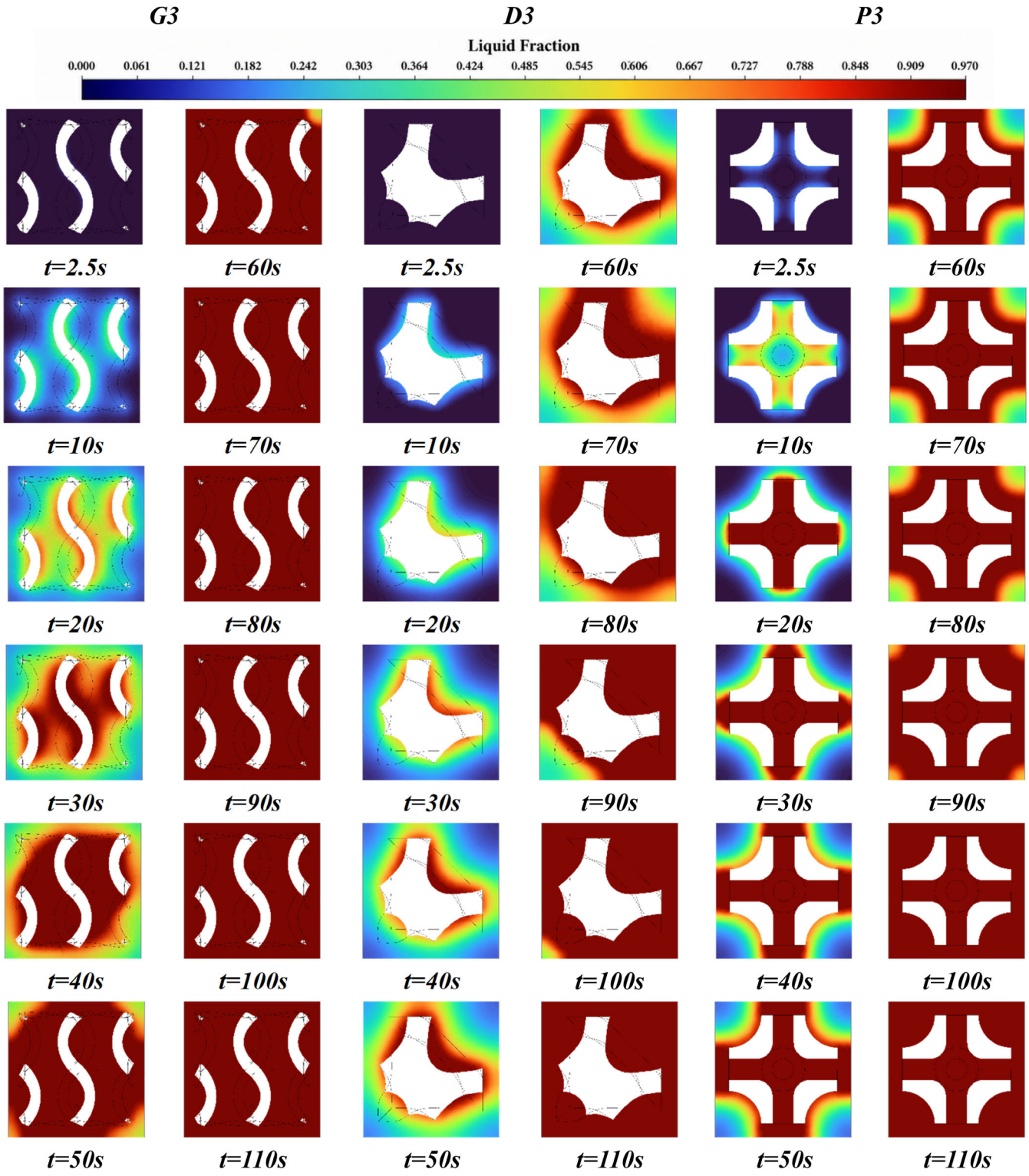



**FIGURE 10** Liquid-fraction contours on the ($XZ$) mid-plane for configurations G3, D3, and P3 at different melting times at the same porosity level of 85.11%.

Therefore, differences among G3, D3, and P3 mainly reflect the effect of TPMS architecture. The 57.29 s melting time is thus specific to the conditions studied and should not be considered universal.

The (XY)-plane contours also help to understand the spatial distribution of melting, as described in Figure 11. Melting is more uniformly distributed across the plane in the Gyroid topology, indicating that heat is transferred more efficiently from the solid matrix to the PCM and thermal energy is disseminated more evenly throughout the domain. G3's faster response can be attributed to a greater interaction between conductive heat spreading and natural convection. The

19447450, 0, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/ep.70734 by Mahidol University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

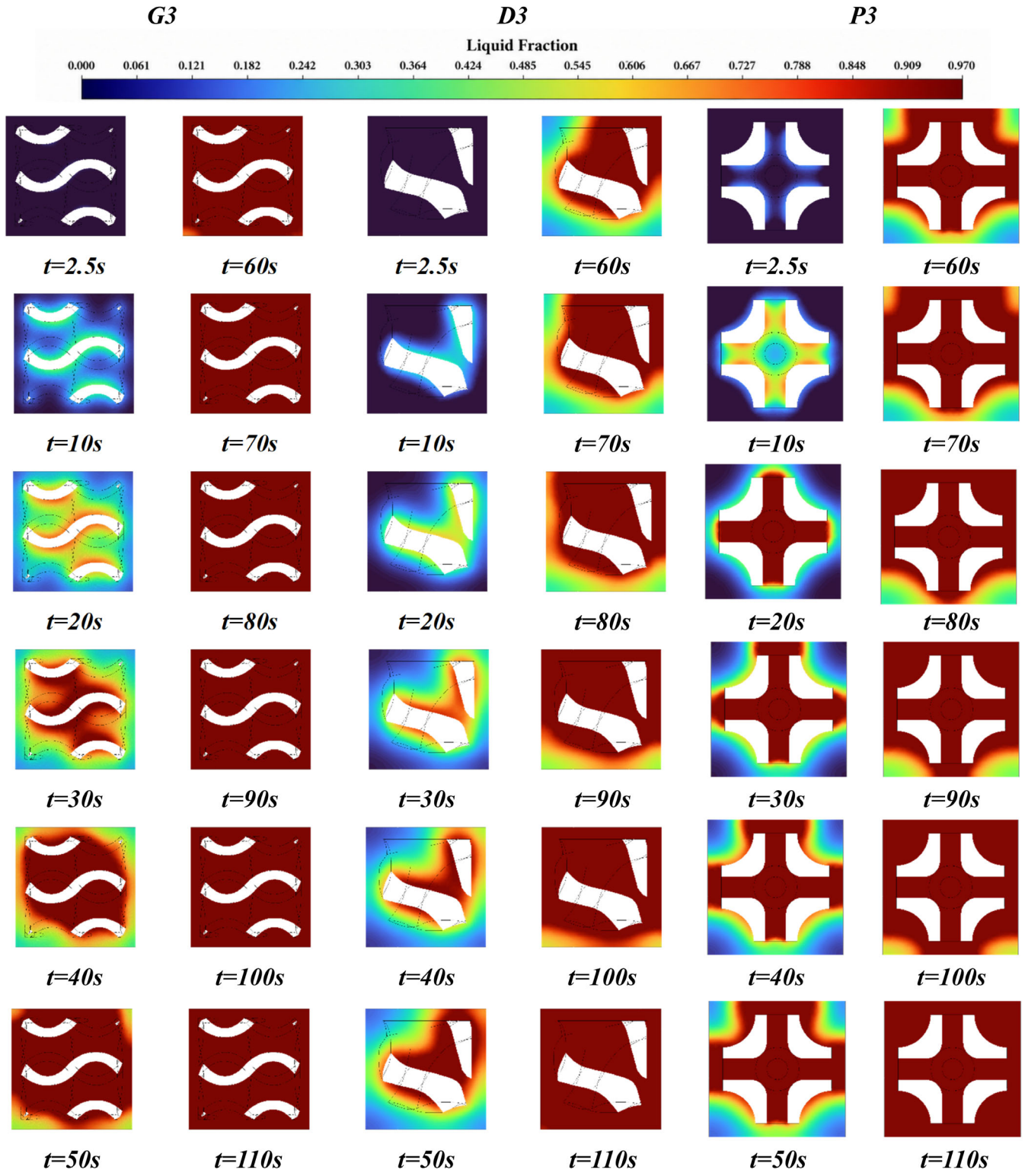



**FIGURE 11** Liquid-fraction contours on the ($XY$) mid-plane for configurations G3, D3, and P3 at different melting times at the same porosity level of 85.11%.

Gyroid structure's continuous curved channels are likely to eliminate stationary zones and improve the circulation of molten PCM. This improves thermal mixing and distributes heat more evenly throughout the PCM volume. In contrast, the Schwarz D and Schwarz P structures appear to delay the formation of effective convective transfer, which justifies their slower melting rate.

In summary, this study shows that TPMS geometry selection is an important design factor for PCM-based thermal energy storage

19447450, 0, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/ep.70734 by Mahidol University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

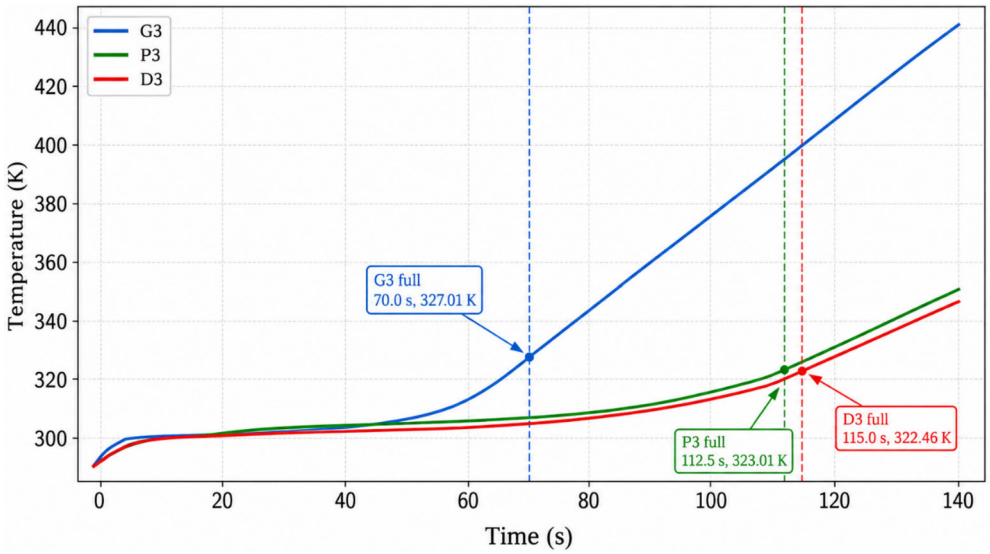



**FIGURE 12** Temporal evolution of the volume-averaged PCM temperature for configurations G3, D3, and P3 at the same porosity level of 85.11%.

systems. At equal porosity, using a Gyroid architecture can reduce charging time by more than 42%, making it more suited for applications that require quick heat storage, such as electronic cooling, solar thermal storage, and transient thermal management systems.

The temperature distribution within the PCM domain is also examined to better understand how these geometries affect heat transfer. This analysis assesses each TPMS structure's ability to minimize the temperature gradient, leading to a more stable and efficient heat transfer process. Figure 12 illustrates the time evolution of the volume-averaged PCM temperature in three TPMS configurations: G3, D3, and P3. During the initial phases, from 0 to 50 s, the three curves remain near the PCM phase-change temperature range, indicating that most of the supplied heat is consumed as latent heat rather than causing a significant temperature increase. This trend shows that the melting process is initially dominated by phase-change energy absorption. Full melting occurs much earlier for the G3 configuration at 70 s, with an average temperature of 327.01 K, while P3 and D3 reach full melting later, at 112.5 and 115 s, with average temperatures of 323.01 and 322.46 K, respectively. Therefore, the total melting time of G3 is reduced by about 37.8% compared with P3 and 39.1% compared with D3. This proves that the Gyroid topology is more efficient at enhancing the charging process than the Schwarz-based designs. After full melting, the temperature of G3 rises rapidly because PCM no longer absorbs heat primarily through latent heat. The supplied energy then heats the liquid PCM, causing the steep temperature rise after 70 s. For D3 and P3, the temperature increase remains slow as part of the PCM remains in phase change for a longer period. Figure 10 then shows that G3 melts faster and that there is a transition from latent heat storage to sensible heat storage, with the PCM becoming fully liquid.

Figure 13 shows the temperature distributions on the (XZ) mid-plane for G3, D3, and P3 at a porosity of 85.11%. It provides spatial evidence of the distinct heat propagation mechanisms induced by each TPMS architecture. In G3, the thermal field rises rapidly along the undulating Gyroid channels, producing a long hot region throughout the initial and intermediate phases. The temperature gradient in G3 becomes much warmer and more uniform than in the other configurations at 60–75 s, confirming the strong thermal connectivity of the Gyroid structure. In D3, the temperature field evolves more slowly and remains localized for a longer time around the heated TPMS surfaces. This indicates that the Schwarz Diamond topology provides more directed, less uniform heat transfer pathways, thereby slowing the homogenization of the temperature field. P3 exhibits a more symmetrical thermal pattern typical of its basic design. However, this symmetry does not necessarily lead to faster melting. The temperature contours show that heat is distributed more uniformly and gradually, with the center area and the neighboring PCM ensuring consistent temperature gradients. Both D3 and P3 show a delay in thermal diffusion compared to G3, which accounts for the longer melting times observed in the global temperature and liquid-fraction curves. The contours in the XZ-plane show that the Gyroid structure enhances the vertical and longitudinal heat propagation, while the Schwarz D and Schwarz P structures generate more localized thermal fields. This confirms that the structure influences both total melting time and the spatial distribution uniformity of the thermal distribution in the PCM domain.

The temperature contours on the XY mid-plane are shown in Figure 14, aiming to analyze the transverse thermal propagation in the PCM domain. In the G3 structure, the temperature field spreads laterally in a continuous and relatively homogeneous way. The undulating Gyroid morphology allows for a smooth distribution of heat across the plane, minimizing isolated cold regions and enhancing the thermal activation of the PCM volume. Conversely, the D3 configuration shows a more irregular transverse temperature distribution. The cool zones are kept for a longer time, while the hot zones remain localized around the TPMS surfaces. This suggests that the Schwarz Diamond topology restricts lateral redistribution of heat in the transverse plane and generates more intense temperature gradients during the melting process. In fact, the PCM temperature rises more rapidly in configuration G3, reaching approximately 327 K by the time the PCM is fully

19447450, 0, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/ep.70734 by Mahidol University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

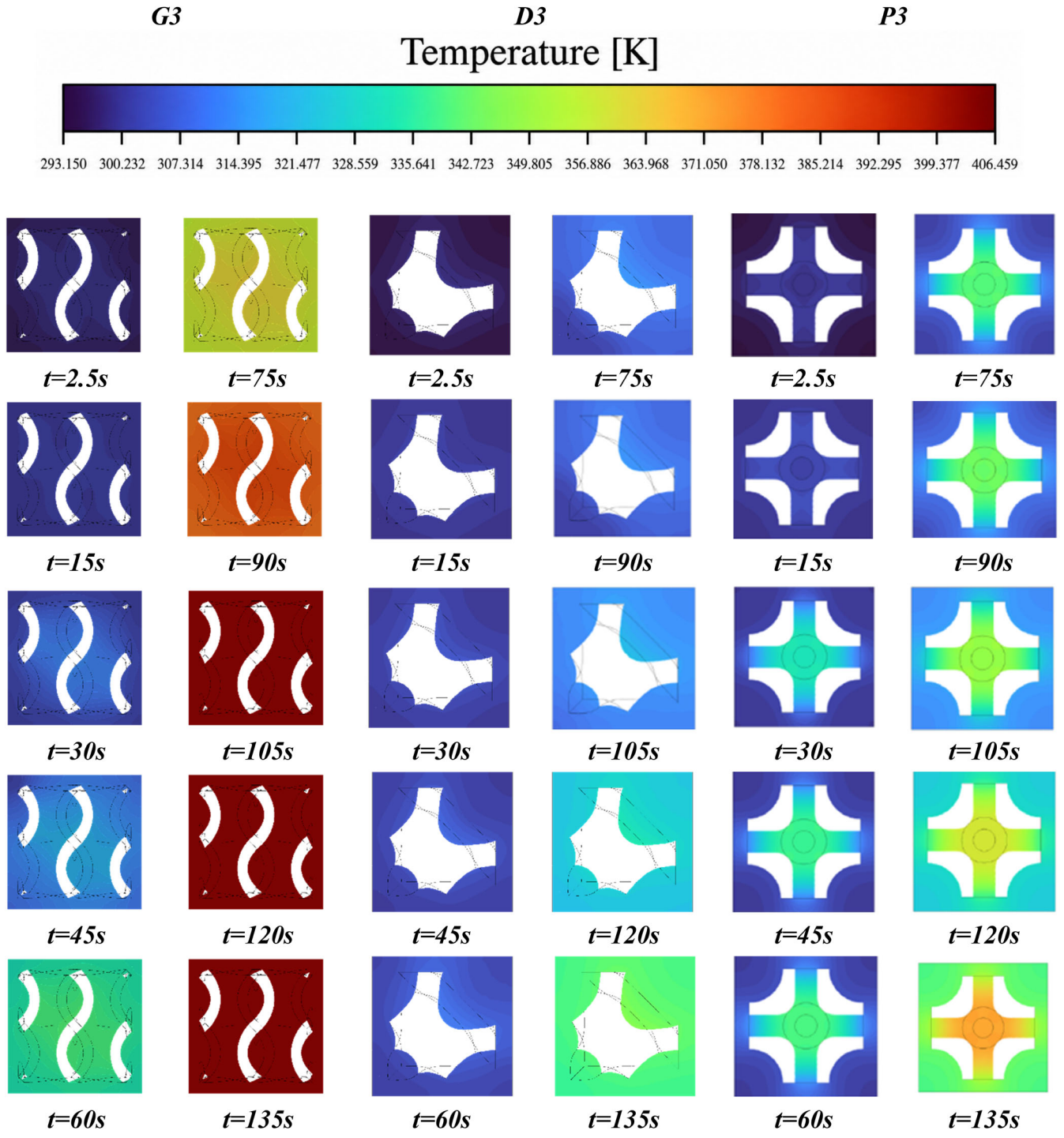



**FIGURE 13** Temperature contours on the $XZ$ mid-plane for configurations G3, D3, and P3 at the same porosity level of 85.11%.

melted. In contrast, in the D3 and P3 configurations, the temperature increase is more gradual, with full melting reached at about 322.46 K after 115 s and 323.01 K after 112.5 s, respectively. It can also be observed that Schwarz P maintains a slightly higher mean temperature than Schwarz D throughout most of the simulation.

From the comparison of Figures 13 and 14, G3 is better in both vertical and transversal planes. Gyroid topology is more effective in promoting three-dimensional thermal spread, while the heat propagation paths of D3 and P3 are more constrained. That's why G3 melts faster and shows better thermal response under the same porosity condition.

Table 4 compares the thermal-uniformity indicators at a common physical time of 90 s. Since the three configurations are at different melting stages at this instant, these values should be interpreted as a common-time comparison rather than a comparison at an identical liquid fraction.

19447450, 0, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/ep.70734 by Mahidol University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

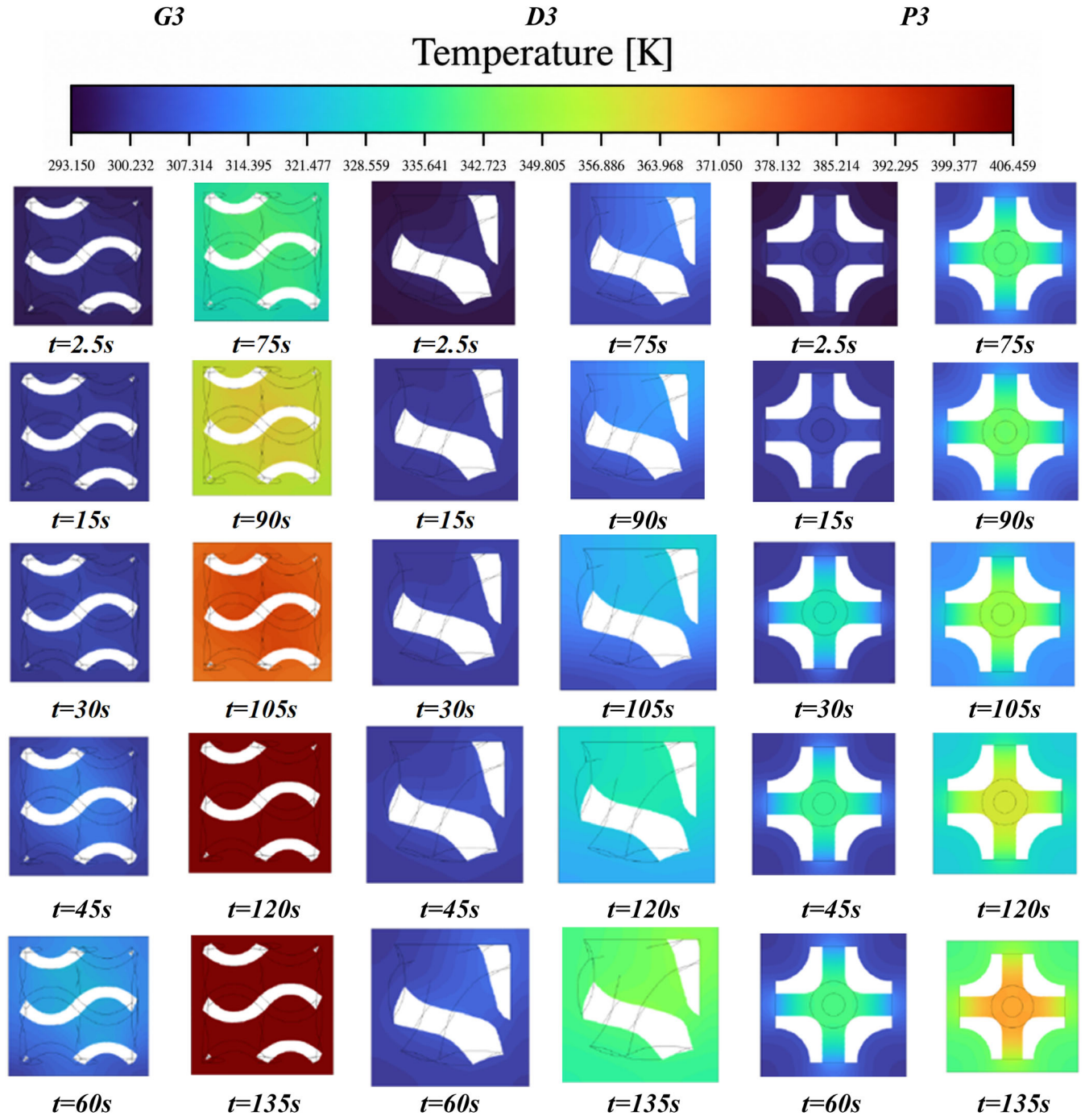



**FIGURE 14** Temperature contours on the $XY$ mid-plane for configurations G3, D3, and P3 at the same porosity level of 85.11%.

**TABLE 4** Quantitative thermal-uniformity indicators at $t = 90$ *s*.

| Configuration | $T_{min}$ (K) | $T_{max}$ (K) | $T_{avg}$ (K) | ΔT(K) | $U_T$(%) |
|---|---|---|---|---|---|
| G3 | ≈355.7 | ≈364.9 | ≈360.2 | ≈9.2 | ≈2.5 |
| D3 | ≈301.6 | ≈320.5 | ≈309.4 | ≈18.9 | ≈6.1 |
| P3 | ≈302.1 | ≈346.8 | ≈310.5 | ≈44.6 | ≈14.4 |

At $t = 90$ s, G3 exhibits the lowest temperature spread ($\Delta T \approx 9.2$ K) and normalized non-uniformity ($U_T \approx 2.5\%$), compared with D3 and P3. These results quantitatively confirm the better thermal uniformity provided by Gyroid topology, as presented in Table 4.

Because the flow inside TPMS-based PCM systems is intrinsically three-dimensional, velocity streamlines were visualized in 3D to better capture the spatial propagation of buoyancy-driven natural convection through the interconnected porous structure. This representation also allows clearer observation of the flow obstruction and recirculation induced by the different TPMS topologies.

Figure 15 displays the velocity streamlines of G3, D3, and P3 at three equivalent melting stages: the beginning of melting λ ≈ 0.2, the

19447450, 0, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/ep.70734 by Mahidol University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

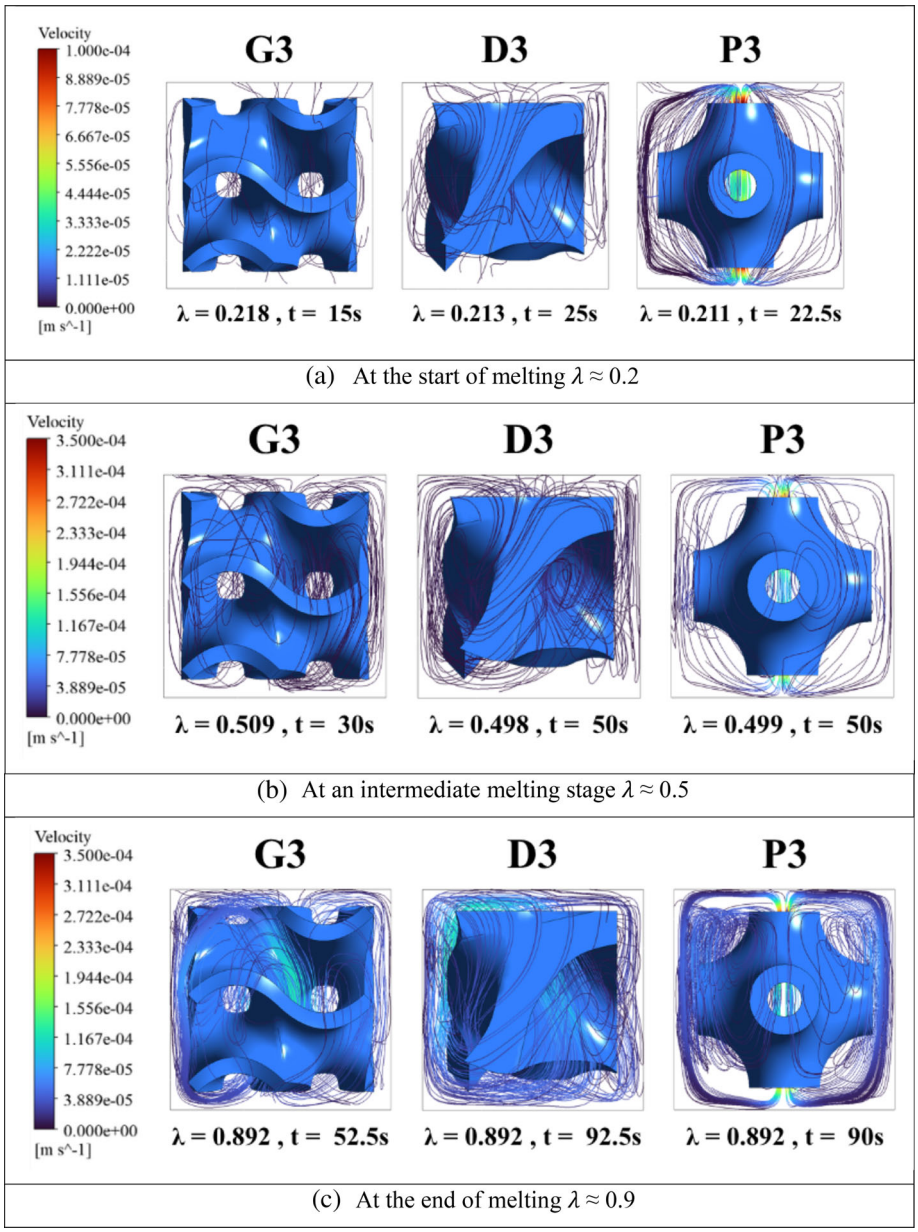



**FIGURE 15** Velocity streamlines for three investigated configurations: Gyroid: G3, Schwarz D: D3, and Schwarz P: P3 at several melting stages approximately λ range of 0.2–0.5–0.9.

intermediate stage $\lambda \approx 0.5$, and the advanced melting stage $\lambda \approx 0.9$. At the beginning of melting, when the liquid fraction $\lambda \approx 0.2$, the flow is weak for all geometries and fluid motion is still at the early stages of development, but the differences in flow organization already start to appear. Configuration G3 converges to $\lambda = 0.218$ in only 15 s at the early stage, while D3 and P3 require 25 and 22.5 s, respectively. This translates to the Gyroid configuration achieving the same initial level of melting approximately 40% faster than D3 and 33% faster than P3. At this stage, the maximum velocity stays low, of the order of $1.0 \times 10^{-4}$ m.s$^{-1}$, demonstrating that natural convection is still weak and conduction is the dominant heat-transfer mechanism. The difference is more pronounced in the intermediate stage. G3 reaches $\lambda = 0.509$ at 30 s, while D3 and P3 reach almost the same liquid fraction only at 50 s. Therefore, the Gyroid structure speeds up the intermediate melting phase by about 40%. The velocity scale reaches about $3.5 \times 10^{-4}$ m.s$^{-1}$, which is almost 3.5 times larger than at the beginning of melting. The increase is an indication of the gradual development of buoyancy-driven natural convection with increasing volume of the liquid PCM. At the intermediate stage of melting, corresponding to $\lambda \approx 0.5$, the flow for the Gyroid configuration seems more distributed throughout the domain, which is consistent with its highly interconnected geometry. By contrast, the flow for the Schwarz D structure exhibits a highly localized but still stronger flow with recirculation zones concentrated in specific regions of the domain. At the advanced melting stage, three forms are compared at almost the same liquid fraction, $\lambda = 0.892$. G3 attains this stage at 52.5 s vs. 92.5 s for D3 and 90 s for P3. Therefore, G3 reduces the time to reach this advanced melting level by about 43.2% and 41.7% compared with D3 and P3, respectively. This demonstrates that the Gyroid topology improves not only the initial melting but the entire phase-change process.

19447450, 0, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/ep.70734 by Mahidol University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

From the flow-structure perspective, G3 shows more evenly distributed and continuous streamline patterns in the PCM domain. This indicates that the Gyroid geometry allows for smoother flow passages and decreases stagnant zones, thus enhancing the interaction between interfacial heat transfer and natural convection in the molten PCM. D3 and P3 instead show more localized recirculation areas and less homogeneous development of the flow, limiting the convective transport and delaying the progression of the melting front.

Generally, the streamline analysis is in agreement with the fact that improved efficiency of the Gyroid configuration is attributed not only to improved conduction but also to its capability to enhance the earlier and more effective buoyancy-driven circulation. This explains the higher melting rates in the liquid fraction and temperature outcomes. Overall, these streamline figures suggest that Gyroid geometry promotes more uniform and distributed fluid motion. For the Schwarz P and D geometries, they both exhibit highly localized flows leading to similar behavior in terms of melting rate and temperature fields.

To assess the relative contributions of convective transport and thermal diffusion in PCM, the Péclet number *Pe* and Prandtl number *Pr* were evaluated. The Péclet number compares advective heat transport with thermal diffusion, while the Prandtl number relates momentum diffusivity to thermal diffusivity.

$$\mathrm{Pe} = \frac{\mathrm{UL}}{\alpha} = \mathrm{Re}\,\mathrm{Pr} \tag{22}$$

With

$$Pr = \frac{\nu}{\alpha} \approx 349.9 \tag{23}$$

where $U$ is the characteristic velocity, $L$ is the characteristic length, $\alpha$ is the thermal diffusivity, and $\nu$ is the kinematic viscosity of RT31.

The progressive increase in *Pe* from approximately 2.5–8.7 indicates that advective heat transport becomes increasingly important relative to thermal diffusion as the liquid PCM volume grows. This quantitative trend supports the transition from an initially conduction-controlled melting process toward a progressively stronger buoyancy-assisted regime. The low Reynolds numbers confirm that this convective flow remains laminar.

The comparison carried out in this section shows that the geometry of the TPMS has a strong effect on the thermal behavior of the PCM system. Among the studied geometries, the Gyroid stands out as the most effective. It exhibits the fastest melting rate and shows a more uniform temperature distribution, together with a highly developed and interconnected flow structure. This can be attributed to its continuous, smooth, and highly interconnected geometry, which promotes fluid circulation, mixing, and rapid heat transfer throughout the domain. By contrast, the Schwarz P and Schwarz D geometries exhibit very similar overall behavior, with the main difference appearing in the flow structure. While both lead to relatively localized recirculation zones, the flow around the Schwarz P geometry is significantly more symmetrical and organized.

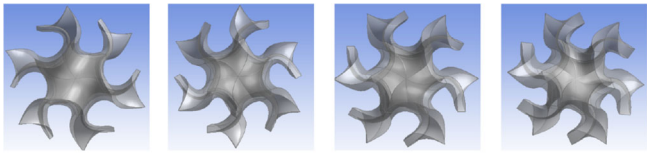



**FIGURE 16** Description of gyroid configurations for different porosities.

The superior performance of the Gyroid cannot be attributed to porosity, since G3, D3, and P3 have the same porosity of 85.11%. The difference therefore originates mainly from the internal architecture of the TPMS. The continuous and smoothly interconnected Gyroid pathways promote a more distributed heat transfer and facilitate the development of buoyancy-driven circulation, whereas the Schwarz structures generate more localized recirculation and stagnant regions. This explains the substantially shorter melting time observed for G3.

## 3.2 | TPMS porosity effect

Porosity is a key design parameter that controls the ratio of conductive heat transfer through the aluminum TPMS skeleton to buoyancy-driven natural convection in the molten PCM. The aim of this analysis is therefore to find a balanced porosity range among the investigated configurations that maximizes melting while maintaining an efficient convective motion. The effect of the porosity of the TPMS on the melting behavior of the PCM is explored in this section. For each geometry, four porosity configurations are examined to determine the effect of this parameter on the melting rate and the evolution of the melting front, the temperature distribution throughout the domain, and the flow structure, as mentioned in Figure 2.

Since the Gyroid is the best configuration, different porosity levels were considered for each topology to evaluate the influence of the amount of metallic structure and volume of PCM on heat transfer, liquid-fraction evolution, and temperature distribution (Figure 16).

Figure 17 illustrates the effect of porosity on the melting behavior of Gyroid-based TPMS/PCM systems. The time to reach $\lambda = 0.95$ increases with porosity; for the Gyroid structure, it is 50.32 s at 78.18% porosity, 57.29 s at 85.11%, 59.78 s at 87.33%, and 65.76 s at 92.03%. Therefore, the lowest-porosity configuration accelerates melting by approximately 12.4%, 16.0%, and 23.7% relative to the 85.11%, 87.33%, and 92.03% cases, respectively. The behavior is primarily attributed to the higher aluminum volume fraction at lower porosity. As porosity decreases, the TPMS–PCM interfacial area increases while the PCM volume decreases, which contributes to faster melting. However, the narrower flow passages may restrict buoyancy-driven convection. This leads to a faster advance of the melting front and to the near-complete melting of the PCM at earlier times. On the other hand, the case with maximum porosity of 92.03% has a larger volume of PCM but a smaller metallic conductive network that increases the thermal resistance and delays the melting process.

19447450, 0, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/ep.70734 by Mahidol University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

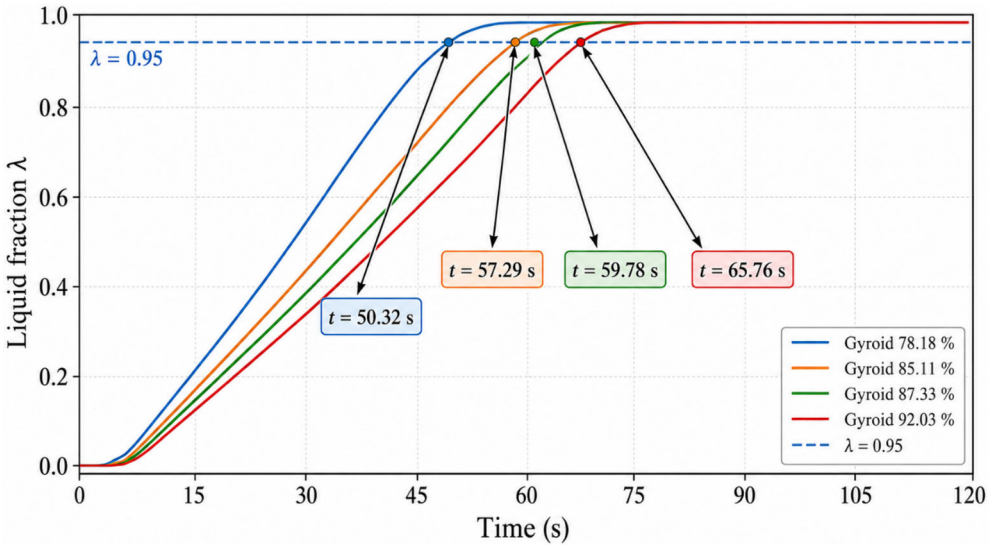



**FIGURE 17** Mean liquid fraction as a function of time for different porosity levels: Gyroid structure.

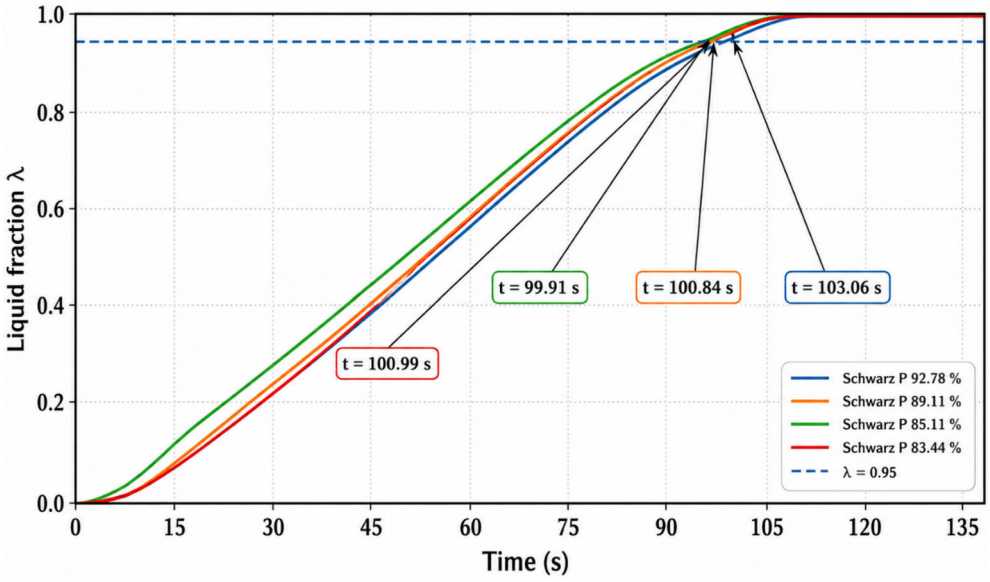



**FIGURE 18** Mean liquid fraction as a function of time for different porosity levels: Schwarz P structure.

However, the fastest melting time does not necessarily lead to the best global thermal-storage design. The case with 78.18% porosity has the least charging time. Higher porosity yields higher PCM volume and storage capacity but less conductive enhancement. So, the intermediate porosities of 85.11% and 87.33% are a more balanced design with faster melting and a larger PCM fraction with better potential energy storage density.

Figure 18 presents the temporal evolution of the mean PCM liquid fraction throughout the simulation for the Schwarz P configurations. For the Schwarz P designs, the influence of porosity on the melting rate is found to be relatively weak compared to the Gyroid case. The time to reach $\lambda = 0.95$ varies from 99.91 s for 85.11% porosity to 103.06 s for 92.78% porosity, with the greatest difference of about 3.1%. The fastest melting is obtained for the intermediate porosity of 85.11%, indicating a better compromise between the heat conduction via the aluminum skeleton and the PCM volume available for phase change. The case of the highest porosity results in a slower melting rate due to the reduced metallic fraction, which reduces the conductive heat transfer. The case of the lowest porosity does not lead to any further improvement, probably due to the reduced PCM volume and the restricted flow paths. Overall, the Schwarz P topology does not show a high sensitivity to porosity since all configurations show similar melting behavior.

Figure 19 shows the effect of porosity on the melting behavior of the TPMS configurations of the Schwarz Diamond. The time to reach $\lambda = 0.95$ decreases from 117.29 s at 90.07% porosity to 109.25 s at 87.74%, 100.89 s at 85.11%, and 82.57 s at 76.30%. Thus, the decrease in porosity from 90.07% to 76.30% decreases the melting time by 34.72 s, which corresponds to an improvement of about 29.6%. This behavior is mainly because of the increase in the volume fraction of aluminum at lower porosity. A denser Schwarz D structure gives more continuous conductive paths to reduce the thermal resistance and accelerate the heat propagation through the PCM. The fastest melting is found for the case with 76.30% porosity, but it

19447450, 0, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/ep.70734 by Mahidol University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

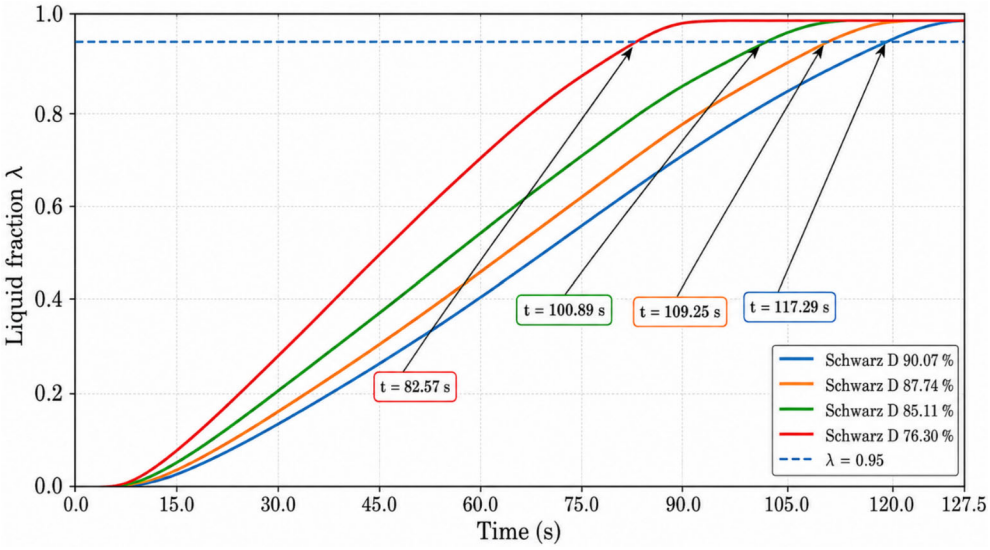



**FIGURE 19** Mean liquid fraction as a function of time for different porosity levels: Schwarz D structure.

**TABLE 5** Latent heat storage capacity of the investigated TPMS configurations.

| Configuration | PCM volume ($mm^3$) | Latent storage capacity $Q_{lat}$(J) |
|---|---|---|
| G1 | 24.848 | 3.403 |
| G2 | 23.579 | 3.229 |
| G3 | 22.980 | 3.147 |
| G4 | 21.109 | 2.891 |
| D1 | 24.319 | 3.330 |
| D2 | 23.690 | 3.244 |
| D3 | 22.980 | 3.147 |
| D4 | 20.601 | 2.821 |
| P1 | 25.051 | 3.431 |
| P2 | 24.060 | 3.295 |
| P3 | 22.980 | 3.147 |
| P4 | 22.529 | 3.085 |

decreases the volume of PCM for latent heat storage and may limit the natural convection. Therefore, the intermediate porosity of 85.11% remains a more balanced structure, combining slower melting with better PCM storage capacity.

The results show that latent heat storage capacity increases with PCM volume and therefore generally increases with porosity (Table 5). For the Gyroid configurations, $Q_{lat}$ decreases from approximately 3.40 J for G1 to 2.89 J for G4, corresponding to a reduction of about 15.0%, although G4 exhibits the shortest melting time. A similar trend is observed for Schwarz Diamond, where the capacity decreases from 3.33 J for D1 to 2.82 J for D4. For Schwarz Primitive, the variation is smaller, from 3.43 J for P1 to 3.09 J for P4. These results quantitatively confirm the trade-off between rapid melting and latent heat storage capacity: lower porosity increases the metallic fraction and accelerates melting but reduces the PCM volume available for latent heat storage. Conversely, higher porosity increases storage capacity but generally results in slower melting. The intermediate-porosity configurations therefore provide a more balanced compromise between charging rate and latent heat storage capacity.

At the common porosity of 85.11%, G3, D3, and P3 have the same PCM volume and consequently the same latent heat storage capacity of approximately 3.15 J, allowing the influence of topology on melting behavior to be compared independently of storage capacity.

The progression of the PCM liquid fraction on the mid-plane XZ for the four Gyroid configurations is depicted in Figure 20. The contours clearly show that the melting process is accelerated with decreasing porosity. The structure with the lowest porosity, G4 with 78.18% porosity, exhibits the fastest diffusion of the phase change, whereby by 30–45 s a large fraction of the PCM is already melted. This behavior is ascribed to the increased aluminum content, which generates strong conductive pathways and better heat penetration into the PCM domain.

On the other hand, the case with the biggest porosity, G1 with 92.03%, presents a slower melting evolution, especially in the intermediate stage. G1 has more PCM and hence more latent storage capacity. But a weaker metallic skeleton that reduces heat. The intermediate structures G2 and G3 show a very similar behavior, which indicates a better balance between the metallic conduction and the PCM storage volume. These intermediate porosities are more interesting for real LHTES systems as they combine fast melting with a high PCM volume.

Figure 21 depicts the evolution of the liquid fraction in the XY mid-plane for the same Gyroid configurations. The results confirm that the melting front develops more rapidly with decreasing porosity. In the G4 configuration, the molten regions are more uniformly distributed in the transverse plane, which indicates that a denser Gyroid structure leads to better lateral heat distribution. This reduces cold regions and allows for a faster transition from partial to almost full melting. The findings confirm that very low porosity increases the melting rate, while intermediate porosities provide the most balanced thermal-storage performances.

19447450, 0, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/ep.70734 by Mahidol University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

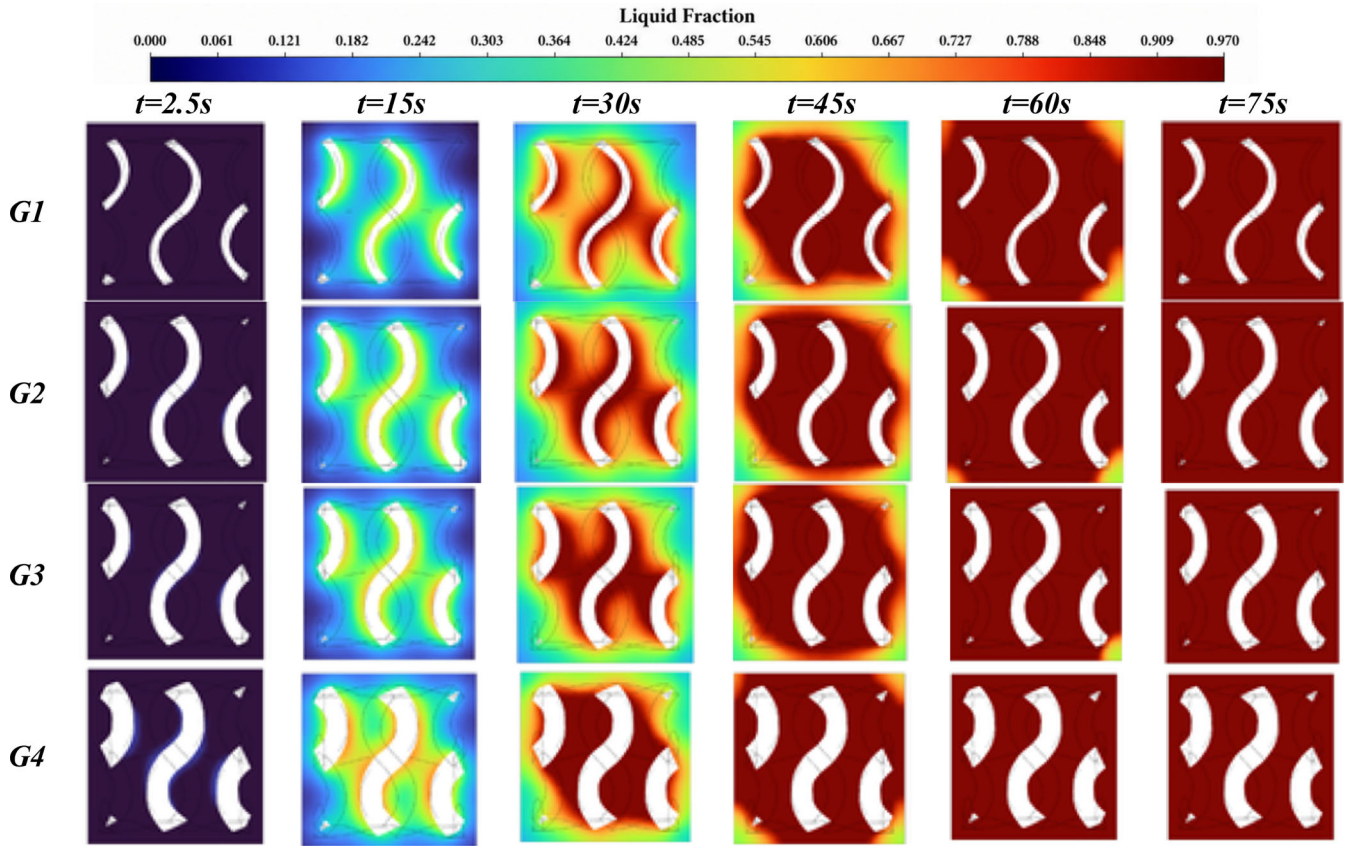



**FIGURE 20** Contours of the PCM liquid fraction at different stages of the melting process for the Gyroid configurations with different porosity levels located in the (XZ) plane.

Temperature contours show that the effect of porosity is not only limited to the melting rate but also affects the overall thermal behavior of the PCM system. Moreover, Figures 22 and 23 show that the dependency of the temperature field on TPMS porosity becomes increasingly substantial as time progresses. Between $t = 2.5$ s and $t = 20$ s, the influence of porosity is not yet clear. The temperature across the domain remains close to the initial temperature for all configurations. Over time, lower-porosity structures start showing faster heat propagation, while higher porosity leads to a less uniform temperature distribution, with hot spots appearing around the TPMS geometry. This effect is even more pronounced at later stages of the phase change. At $t = 70$ s, the G4 configuration shows a more uniform temperature distribution and reaches higher temperatures, around 350 K, while the G1 configuration still contains colder regions with temperatures around 305 K. This suggests that highly porous structures result in larger temperature gradients across the PCM domain, whereas less porous structures can distribute heat more evenly and efficiently.

The previous results showed that lower-porosity TPMS structures result in faster melting of the PCM and a more uniform heat distribution. However, these results are not sufficient to conclude on the effect of TPMS porosity on heat transfer within the PCM, which is controlled by two main mechanisms: conduction and natural convection. To assess this effect, the flow structure must be studied to determine the influence of TPMS porosity on the buoyancy-driven flow that develops as the PCM melts.

At a more advanced stage of the melting process, corresponding to $\lambda \approx 0.95\%$, natural convection becomes fully developed, as described in Figure 24. This is reflected in the denser streamlines, higher velocity magnitudes, and larger recirculation zones within the PCM domain. Buoyancy-driven flow plays a more important role at this stage compared with the early and intermediate stages of the phase change. Among the different configurations, the 92.03% porosity Gyroid shows the most developed flow structure. Its lower solid fraction provides larger flow passages, allowing the liquid PCM to circulate more freely around the Gyroid pores and promoting stronger natural convection. The G2 and G3 configurations also exhibit strong convection-driven heat transfer despite less developed flow structures than the G1. By contrast, the 78.18% Gyroid configuration leads to the weakest fluid circulation. It shows fewer recirculation paths and lower velocity magnitudes. This suggests that, despite its faster melting rate, the reduced PCM volume and more confined flow passages considerably limit the development of natural convection.

The 3D streamlines show that the Gyroid configuration promotes more continuous and spatially distributed flow paths, indicating better propagation of natural convection through the PCM domain. In contrast, the Schwarz Diamond and Schwarz Primitive structures

19447450, 0, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/ep.70734 by Mahidol University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

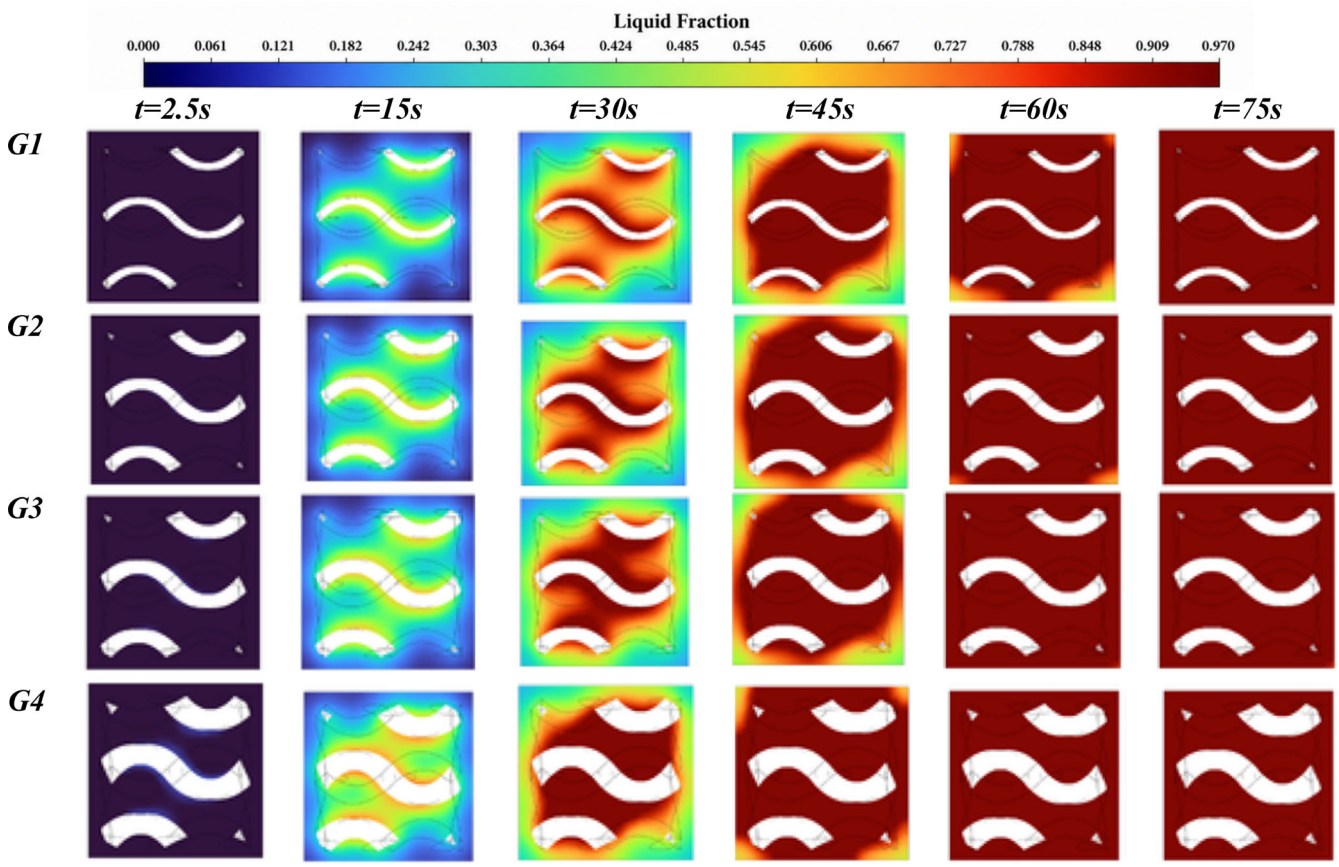



**FIGURE 21** Contours of the PCM liquid fraction at different stages of the melting process for the Gyroid configurations with different porosity levels located in the (XY) plane.

generate more localized recirculation zones and more restricted flow passages, revealing a stronger barrier effect to buoyancy-driven motion. This behavior is consistent with their slower melting response.

The previous comparison shows that lower porosity leads to faster PCM melting and a more uniform temperature distribution. However, when examining the flow structure of the PCM at different stages of melting, regardless of the time required to reach each stage, it can be concluded that if the porosity is too low, the TPMS structure interferes with the development of natural convection within the PCM domain and slows down buoyancy-driven fluid motion. This becomes particularly apparent toward the end of melting, where the role of natural convection in heat transfer is more substantial.

Therefore, porosity is a particularly important geometrical parameter for TPMS structures. By controlling the PCM-to-metal volume ratio, it directly affects the effective thermal conductivity of the system and the relative contribution of natural convection. As shown in this section, increasing the amount of solid TPMS within the PCM system can lead to faster melting. This was reflected by the higher initial slope of the liquid fraction curve and the faster propagation of the melting front. A lower porosity also promotes a more uniform temperature distribution within the PCM domain, with less pronounced temperature gradients. However, the results also showed that reducing the porosity limits the development of fluid flow inside the PCM. The streamline contours revealed a weaker and less organized flow structure for the lower-porosity configurations. This indicates that the reduced pore space limits the available pathways for fluid circulation, thereby restricting buoyancy-driven motion.

The intermediate porosity range was not obtained from a formal optimization algorithm. It was identified from a multi-criteria comparison considering melting time, PCM storage volume, thermal uniformity, and natural-convection development. Lower porosity accelerates melting but reduces PCM volume and restricts convection, whereas higher porosity increases PCM capacity and fluid circulation but weakens conductive heat transfer. Therefore, intermediate porosities represent the most balanced design among the investigated cases.

Results show that Gyroid topology offers the most effective thermal enhancement, leading to faster melting and a more uniform temperature field than Schwarz Diamond and Schwarz Primitive configurations. The enhancement in performance is due to the advantages of the continuous interconnected geometry for heat diffusion and buoyancy-driven circulation. The porosity analysis indicates that very low porosity is conducive to conduction but

19447450, 0, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/ep.70734 by Mahidol University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

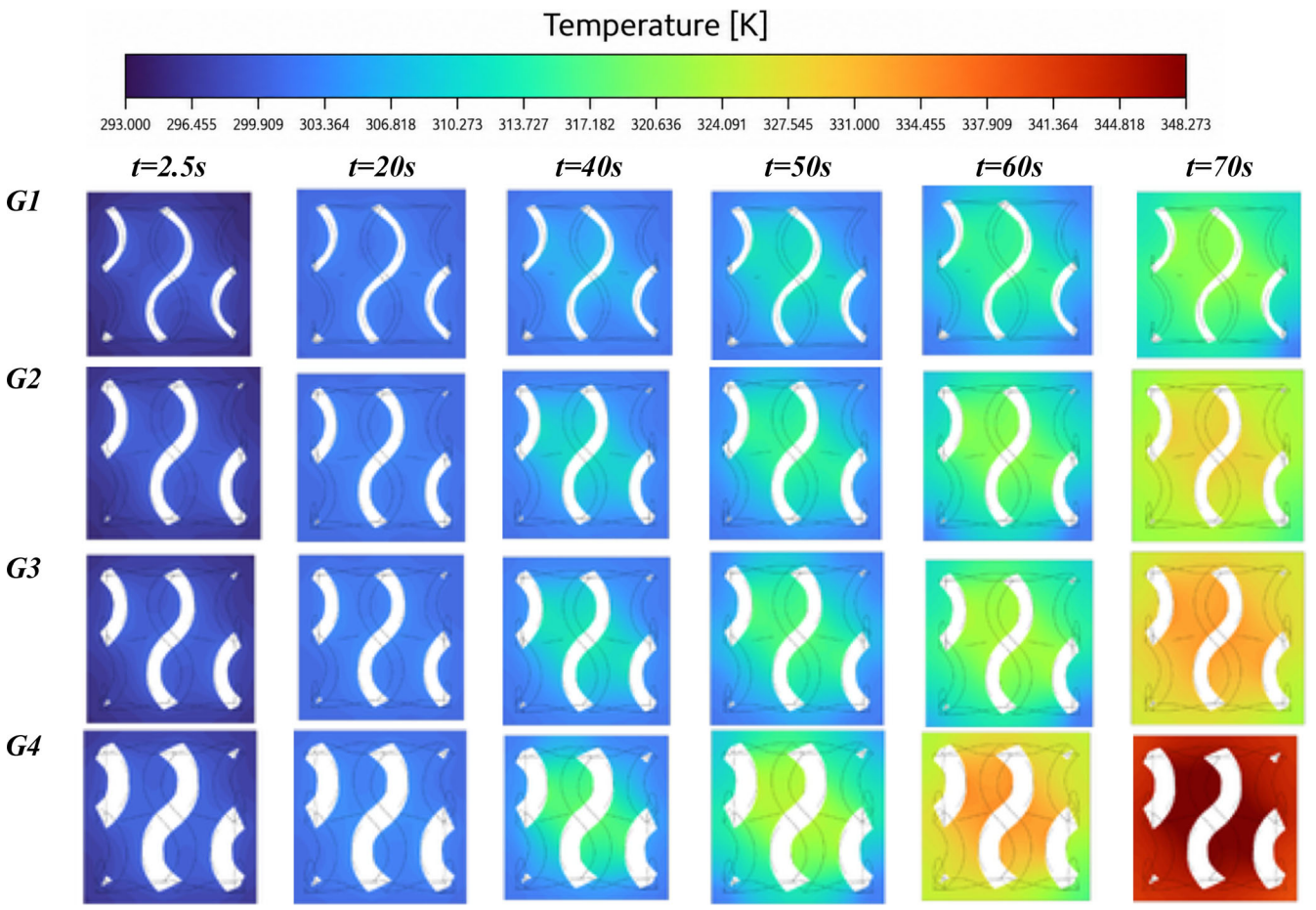



**FIGURE 22** Contours of the PCM temperature at different stages of the melting process for the Gyroid configurations with different porosity levels located in the (XZ) plane.

may suppress natural convection, while very high porosity allows for fluid motion but reduces the conductive solid fraction. Thus, intermediate Gyroid porosities of 85.11% and 87.33% provide a balanced trade-off among the investigated configurations between conductive enhancement and convective transport. The results demonstrate the importance of a combined topology-porosity design strategy for improving the performance of PCM-based latent heat storage systems.

The porosity results reveal a clear competition between conduction, natural convection, and storage capacity. Reducing porosity increases the solid fraction and accelerates melting but simultaneously decreases the PCM volume and restricts buoyancy-driven flow. Conversely, higher porosity provides more PCM storage volume and wider fluid passages but weakens the conductive contribution. Therefore, the fastest-melting configuration is not necessarily the most balanced thermal-storage design.

To quantitatively evaluate the thermal performance of the different TPMS configurations, three performance indicators were considered: the latent energy storage capacity per unit system mass, the latent energy storage capacity per unit system volume, and the average charging power required to reach 95% melting.

- $E_m$: latent energy storage per unit total system mass

$$E_m = \frac{Q_{lat}}{m_{PCM} + m_{Al}} \tag{24}$$

- $E_v$: latent energy storage per unit total system volume

$$E_v = \frac{Q_{lat}}{V_{system}} \tag{25}$$

- $P_{95}$: average latent charging power required to reach 95% melting

The normalized results confirm the competition between charging rate and storage density. For example, within the Gyroid family, decreasing porosity from G1 to G4 increases the average melting rate from $1.45 \times 10^{-2} s^{-1}$ to $1.89 \times 10^{-2} s^{-1}$, while the specific latent storage capacity decreases from approximately 128.5–86.2 kJ $kg^{-1}$ and the volumetric capacity decreases from 126.0 to 107.1 MJ $m^{-3}$. Therefore, the lowest-porosity structure provides the fastest charging but not the highest storage density, quantitatively supporting the selection of intermediate porosity as a more balanced design (Table 6).

19447450, 0, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/ep.70734 by Mahidol University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

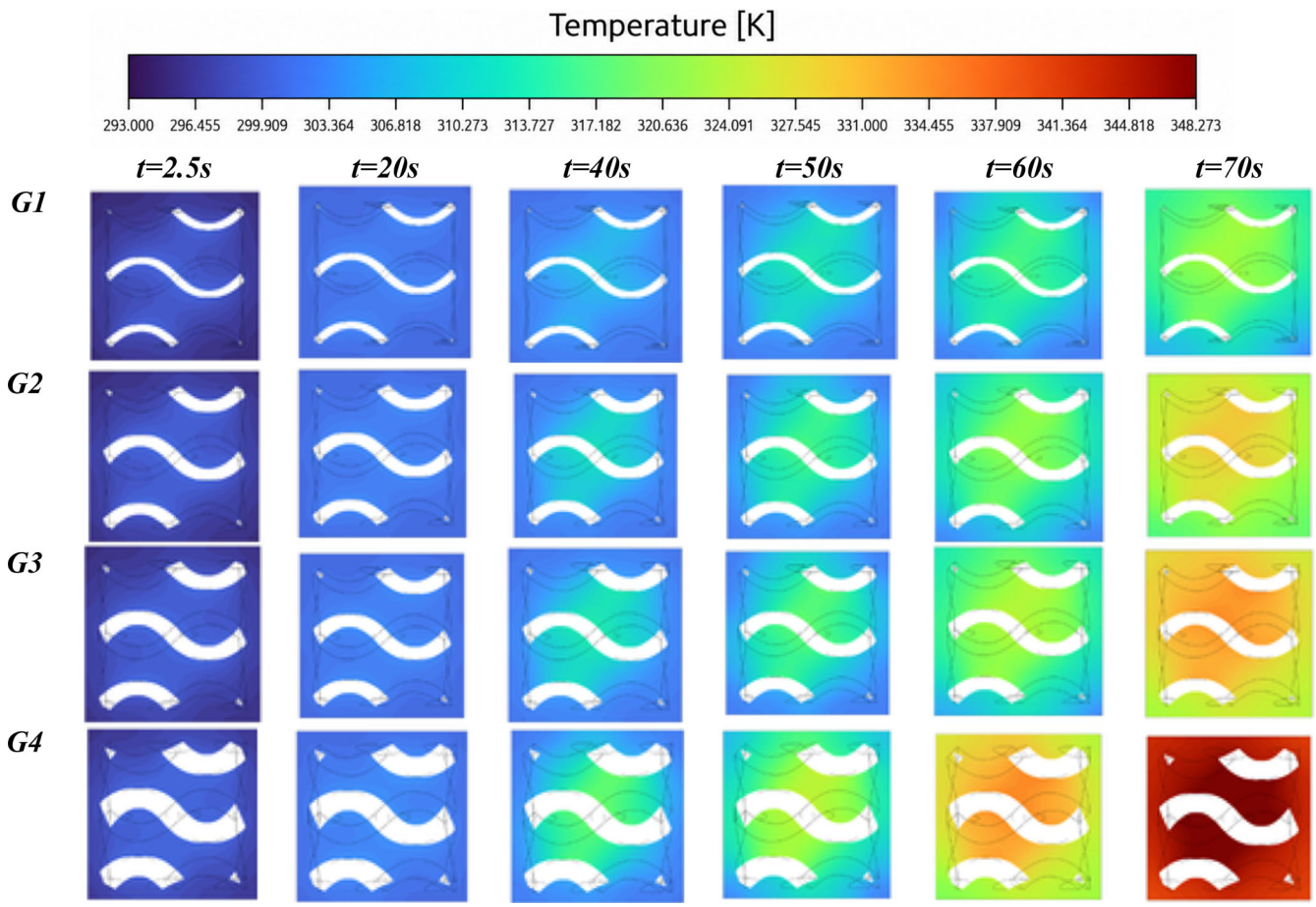



**FIGURE 23** Contours of PCM temperature at different stages of the melting process for the Gyroid configurations with different porosity levels located in the (XY) plane.

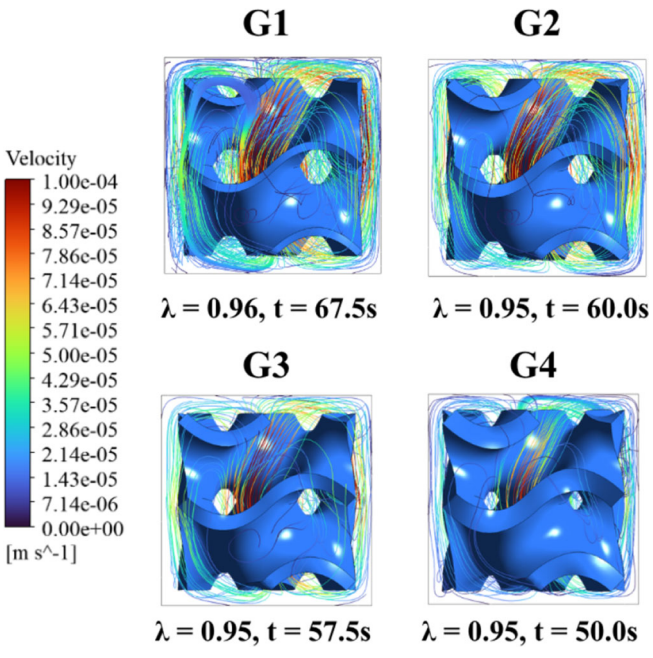



**FIGURE 24** Velocity streamlines for the Gyroid configurations with different porosity levels near the end of melting $\lambda \approx 0.95$.

### 3.3 | Comparative discussion with previous literature

To further highlight the contribution of the present work, the current results are compared with previous studies on TPMS-based thermal enhancement systems in Table 7. Previous works mainly showed the potential of TPMS structures to improve melting, electronic cooling, or the performance of compact heat exchangers.[21] The present study is devoted to the coupled effect of topology and porosity under the same thermal and numerical conditions.

The present results are consistent with previous studies showing that TPMS architecture can strongly enhance PCM melting by modifying both conductive heat transfer and buoyancy-driven flow. Qureshi et al.[21] reported a 31%–40% reduction in melting time for TPMS-based structures compared with Kelvin-cell foam, confirming that thermal performance depends not only on porosity but also on the topology of the conductive network. Similarly, Nie et al.[22] found melting-time reductions of 47%–75.5% for Diamond-based structures compared with conventional foam configurations. In the present study, the Gyroid structure reaches 95% melting in 57.29 s, approximately 43% faster than the Schwarz Diamond and Schwarz Primitive configurations at the same porosity of 85.11%. This result confirms

19447450, 0, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/ep.70734 by Mahidol University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

**TABLE 6** Latent heat storage and melting performance of TPMS configurations.

| Config. | $Q_{lat}$(J) | $E_m$(kJ $kg^{-1}$) | $E_v$(MJ $m^{-3}$) | $t_{95}$(s) | Avg. melting rate 0.95/$t_{95}$($s^{-1}$) |
|---|---|---|---|---|---|
| G1 | 3.403 | 128.5 | 126.0 | 65.76 | 0.01445 |
| G2 | 3.229 | 111.8 | 119.6 | 59.78 | 0.01589 |
| G3 | 3.147 | 104.9 | 116.6 | 57.29 | 0.01658 |
| G4 | 2.891 | 86.2 | 107.1 | 50.32 | 0.01888 |
| D1 | 3.330 | 121.2 | 123.4 | 117.29 | 0.00810 |
| D2 | 3.244 | 113.2 | 120.2 | 109.25 | 0.00870 |
| D3 | 3.147 | 104.9 | 116.6 | 100.89 | 0.00942 |
| D4 | 2.821 | 81.8 | 104.5 | 82.57 | 0.01151 |
| P1 | 3.431 | 131.5 | 127.1 | 103.06 | 0.00922 |
| P2 | 3.295 | 117.8 | 122.0 | 100.84 | 0.00942 |
| P3 | 3.147 | 104.9 | 116.6 | 99.91 | 0.00951 |
| P4 | 3.085 | 100.0 | 114.3 | 100.99 | 0.00940 |

**TABLE 7** Comparison of previous TPMS-based thermal enhancement studies with the present work.[21]

| Study | Application | TPMS design | Method | Main findings |
|---|---|---|---|---|
| Qureshi et al.[21] | PCM thermal energy storage | Gyroid, IWP, Primitive | Numerical/ validated model | TPMS structures reduced melting time by about 31%–40% compared with Kelvin-cell foam. |
| Nie et al.[22] | PCM melting in rectangular enclosure | Different TPMS structures, including Diamond | Numerical CFD | Diamond structures reduced melting time by 47%–75.5% and increased power density by 178%–275% compared with Kelvin foam. |
| Zhang et al.[28] | Compact electronic thermal management | TPMS-based PCM capsules | Numerical/design evaluation | TPMS capsules reduced melting time by more than 70% and increased peak charging power by up to 97.2%. |
| Yan et al.[21] | Latent heat thermal storage | Diamond, Primitive, Gyroid; uniform and graded porosity | Numerical + visualization experiments | Diamond showed the best heat-transfer performance; positive porosity grading reduced the time to reach the phase-change temperature by up to 20.5%. Experimentally, TPMS reduced paraffin melting time from 2179 s to 320 s. |
| Su et al.[37] | Functionally graded TPMS–PCM storage | Functionally graded TPMS; different gradient functions and steepness | Numerical | An exponential gradient reduced the time to 50% melting by 17.5%, while a quadratic gradient reduced complete melting time by 6.5%. The uniform structure provided the best temperature uniformity. |
| Huang et al.[27] | Latent heat thermal energy storage | Gyroid, Diamond, Primitive; field-driven porosity | Numerical + visualization experiments | Temperature-field-driven porosity redistribution reduced phase-change time by 68.4 s, 81 s, and 44 s for Gyroid, Diamond, and Primitive, respectively, while simultaneously increasing heat-storage capacity. |
| Zhang et al.[38] | TPMS-based latent heat storage | Topology-optimized TPMS porosity distribution | Transient numerical topology optimization | The topology-optimized TPMS melted 17.4% faster and stored 4.1% more heat than conventional designs, demonstrating the benefit of spatial porosity optimization. |
| Torri et al.[28] | PCM–water compact heat exchanger | Additively manufactured aluminum Gyroid; porosity 0.70–0.80 | Experimental + numerical | Experiments confirmed that TPMS porosity strongly influences phase-change heat transfer; aluminum Gyroid heat exchangers were successfully fabricated and tested. |
| Chtourou et al.[17] | Compact plate heat exchanger | Gyroid | Numerical CFD | Gyroid increased the Nusselt number by 35.24% compared with a smooth channel and improved overall performance by 16.85% relative to a chevron plate heat exchanger. |
| Present study | RT31 latent heat thermal storage | Gyroid, Schwarz Diamond, Schwarz Primitive; multiple porosity levels | 3D CFD, enthalpy–porosity model | Twelve configurations are assessed. At the common porosity of 85.11%, Gyroid reaches 95% melting in 57.29 s, approximately 43% faster than Schwarz Diamond and Schwarz Primitive. The analysis additionally considers thermal uniformity, natural convection, and latent heat storage capacity. |

19447450, 0, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/ep.70734 by Mahidol University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

that topology alone can significantly modify the phase-change dynamics when the PCM volume fraction is kept constant.

## 4 | CONCLUSION

The thermal performance of RT31 PCM enhanced with Gyroid, Schwarz Diamond, and Schwarz Primitive TPMS structures was investigated numerically under identical thermal conditions. At the common porosity of 85.11%, the Gyroid topology showed the fastest melting response, reaching a liquid fraction of 0.95 in 57.29 s, whereas Schwarz Diamond and Schwarz Primitive required 100.89 s and 99.91 s, respectively. This corresponds to melting-time reductions of approximately 43.2% and 42.7%. Complete melting was also achieved earlier for Gyroid, at approximately 70 s, compared with 115 s for Schwarz Diamond and 112.5 s for Schwarz Primitive.

Porosity was also shown to strongly affect the thermal response. For the Gyroid structures, the time required to reach 95% melting decreased from 65.76 s at 92.03% porosity to 50.32 s at 78.18% porosity, corresponding to an improvement of about 23.5%. For Schwarz Diamond, reducing porosity from 90.07% to 76.30% decreased the melting time from 117.29 to 82.57 s, an improvement of approximately 29.6%. By contrast, Schwarz Primitive showed much weaker sensitivity to porosity, with melting times varying by only about 3.1%.

These results demonstrate that lower porosity generally increases the heated TPMS–PCM interfacial area and accelerates melting but simultaneously reduces PCM storage volume and restricts buoyancy-driven circulation. Higher porosity favors natural convection and increases latent storage capacity but weakens the conductive contribution. Therefore, the fastest-melting configuration is not necessarily the most balanced thermal-storage design. Among the investigated cases, intermediate-porosity Gyroid structures provide a balanced trade-off among the investigated configurations between melting rate, thermal uniformity, natural convection, and latent heat storage capacity.

In general, the results emphasize that the design of TPMS-based PCM systems should not be based on the maximization of metallic content or porosity alone. The topology and porosity should be considered jointly to achieve an effective trade-off between rapid melting, thermal uniformity, natural-convection development, and latent heat storage capacity. The most promising design for PCM-based latent heat thermal energy storage systems among the investigated cases is the Gyroid topology with intermediate porosity. The results can be used as useful design guidelines for the application of TPMS-enhanced PCM systems in compact thermal energy storage systems.

A limitation of the present study is the absence of direct experimental validation using fabricated TPMS–RT31 samples. Future work will therefore focus on additive manufacturing of the selected TPMS structures and experimental characterization of PCM melting to validate the predicted melting times, temperature fields, and natural-convection behavior.

## AUTHOR CONTRIBUTIONS

**Sirine Chtourou:** Conceptualization; supervision; data curation; writing – review and editing; writing – original draft. **Ines Ben Ali:** Investigation; funding acquisition; validation; methodology; software; conceptualization. **Lina Chelbi:** Writing – original draft; writing – review and editing; visualization. **Mounir Baccar:** Supervision; project administration; writing – review and editing; software; data curation.

## ACKNOWLEDGMENTS

I would like to express my deep gratitude to the entire staff of the Advanced Fluid Dynamics, Energy, and Environment Laboratory for their scientific contributions and ongoing support in the completion of this work. I also thank the Tunisian Ministry of Higher Education and Scientific Research, the National School of Engineers of Tunis (ENIT), and the National School of Engineers of Sfax (ENIS) for their support and the resources provided that made this project possible.

We confirm that our submission adheres to the journal's AI policy. ChatGPT (OpenAI) was used only for language editing, and no AI-generated scientific content is included.

## DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

## ORCID

*Sirine Chtourou* https://orcid.org/0000-0001-6720-6271
*Lina Chelbi* https://orcid.org/0009-0003-7199-0830

## REFERENCES

1. Showole A, Afaqui MS. Optimization of renewable energy management using machine learning and genetic algorithms. *Energy Storage Sav*. 2026;5(2):100162. doi:10.1016/j.enss.2025.06.007
2. Singh SD, Anand P, Kushwaha P, Roy SN, Chauhan AS. Advancing sustainability through waste heat recovery: a comprehensive review of recovery technologies, applications, and carbon footprint reduction strategies. *Circ Econ Sustain*. 2026;6:171. doi:10.1007/s43615-026-00916-0
3. Khan SA, Abdellatif HE, Belaadi A, Liu H. Computational and machine learning analysis on bio-inspired coronary fin designs for enhanced melting in PCM-based next-generation thermal energy storage system. *Int Commun Heat Mass Transfer*. 2026;170:109925. doi:10.1016/j.icheatmasstransfer.2025.109925
4. Triki R, Chtourou S, Baccar M. Heat transfer enhancement of phase change materials PCMs using innovative fractal H-shaped fin configurations. *J Energy Storage*. 2023;73:109020. doi:10.1016/j.est.2023.109020
5. Zinnah MA, Hassan N, Rasul M, de Silva S. A comprehensive review of the numerical work of the thermal performance of nanoparticle-augmented PCM-based solar thermal energy storage (STES) systems. *J Therm Anal Calorim*. 2026;151:9463-9497. doi:10.1007/s10973-026-15685-z
6. Tat SA, Muthukumar P, Mondal PK. Design, development and performance investigations of a latent heat storage with PCM encapsulation. *J Energy Storage*. 2023;72:108695. doi:10.1016/j.est.2023.108695
7. Dhabarde S, Laad M. Development of hybrid graphene–biochar–metal foam composites phase change materials for building applications. *Mater Renew Sustain Energy*. 2026;15(1):24. doi:10.1007/s40243-026-00365-7

19447450, 0, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/ep.70734 by Mahidol University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

8. Es-satte M, Afaynou I, Faraji H, et al. Phase change materials, bionic structures and artificial intelligence for lithium-ion battery thermal management: a comprehensive review. *Int Commun Heat Mass Transfer*. 2026;173:110870.
9. Said Z, Pandey AK, Tiwari AK, et al. Nano-enhanced phase change materials: fundamentals and applications. *Prog Energy Combust Sci*. 2024;104:101162.
10. Afaynou I, Faraji H, Choukairy K, Djebali R. Metal foam reinforced phase change material for passive thermal control of multiple electronic components. *Int J Therm Sci*. 2026;220:110329. doi:10.1016/j.ijthermalsci.2025.110329
11. Govindaraman LT, Arjunan A, Baroutaji A, Robinson J, Olabi AG. Metamaterials for energy harvesting. *Encyclopedia of Smart Materials*. Elsevier; 2022:522-534. doi:10.1016/B978-0-12-815732-9.00127-3
12. Boadu KG, Adido B, Wu Q. Tubular lattice structures for energy absorption: a review of design, optimization and applications. *Eur J Appl Sci Eng Technol*. 2026;4(2):279-311.
13. Nguyen TT, Le BA, Ţălu Ş. Triply periodic minimal surfaces structures in energy absorption: a review of geometric topologies, manufacturing constraints, and future smart materials. *J Nanomater Appl*. 2026;2(2):16-37. doi:10.65273/hhit.jna.2026.2.2.035
14. Ahmad SZ, Masood MH, Khattab MU, Ahmad SS, Zaidi SAA, Khan SZ. Recent developments in novel TPMS lattice materials: design optimization, performance control, and applications in biomimetic scaffolds. *Materials*. 2025;18(22):5209.
15. Ejeh CJ, Shittu RA, Al-Kaabi AK, et al. Ultra-tough interpenetrating phase cementitious composites with functionally-graded and surface perturbed polymer TPMS lattices: flexural and compressive properties. *Constr Build Mater*. 2026;532:146726.
16. Iqbal F, Sajid M. Triply periodic minimal surface (TPMS) architectures for advanced battery thermal management systems in EVs: manufacturing, integration, and future prospects. *Green Energy Intell Transp*. 2026;5(5):100406. doi:10.1016/j.geits.2026.100406
17. Chtourou S, Hammami S, Hamden J, et al. Heat transfer enhancement in a gyroid-structured compact heat exchanger using computational fluid dynamics. *Int J Eng Res Afr*. 2026;80:59-80. doi:10.4028/p-8Yfn9E
18. Torri F, Berni F, Aider Y, Singh P. Gyroid-based PCM-water compact heat exchangers. *Appl Therm Eng*. 2026;292:130352. doi:10.1016/j.applthermaleng.2026.130352
19. Liu B, Zhu H, Wang Z, et al. Customizing thermal regulation of composite phase change materials using triply periodic minimal surfaces: a coupled simulation and intelligent algorithms optimization framework. *Energy Convers Manag*. 2026;354:121200.
20. Yan K, Liang J, Yu R, Zhou T, Feng Y, Qiu L. Heat transfer tuning-oriented optimization and screening of triply periodic minimal surface structure-phase change material composites. *Appl Therm Eng*. 2026;292:130405. doi:10.1016/j.applthermaleng.2026.130405
21. Qureshi ZA, Al-Omari SAB, Elnajjar E, et al. Using triply periodic minimal surfaces (TPMS)-based metal foam structures as skeletons for metal-foam-PCM composites for thermal energy storage and energy management applications. *Int Commun Heat Mass Transfer*. 2021;124:105265.
22. Nie C, Wu Z, Li H, Liu X, Liu J, Rao Z. Performance comparison of phase change material melting in rectangular container with different triply periodic minimal surface structures. *Int J Therm Sci*. 2026;225:110784.
23. Gado MG. Thermal management and heat transfer enhancement of electronic devices using integrative phase change material (PCM) and triply periodic minimal surface (TPMS) heat sinks. *Appl Therm Eng*. 2025;258:124504.
24. Ghafoorian F, Wang Y, Wan H. Numerical analysis of thermal performance in triply periodic minimal surface structures with phase change material for passive thermal management. *J Energy Storage*. 2025;134:118093.
25. Al-Safadi M, Shuja SZ, Zubair SM. CFD comparative analysis of conventional and triply periodic minimal surface (TPMS) metal foam composites for latent heat thermal energy storage. *Arab J Sci Eng*. 2026;51(1):1023-1042.
26. Yang H, Wang Z, Bao R, Zheng Z, Hong S, Fan Y. Melting characteristics of PCM embedded in TPMS lattices at different inclination angles. *Appl Therm Eng*. 2026;290:130050.
27. Huang W, Tan X, Lv Z, Xia Y, Wu C, Shi J. Triply periodic minimal surface (TPMS) structures with field-driven design to enhance thermal performance of latent heat energy storage system. *Int Commun Heat Mass Transfer*. 2026;176:111316. doi:10.1016/j.icheatmasstransfer.2026.111316
28. Zhang X, Zhang Z, Zhang J, et al. TPMS-based PCM capsules for thermal management in compact electronic systems: design and evaluation. *Energy*. 2025;343:139747.
29. Rubitherm Technologies GmbH. Technical Data Sheet: RUBITHERM® RT31. Berlin, Germany: Rubitherm Technologies GmbH; Version 18.04.2024. 2024.
30. Zhou L, Zhang L, Gao Y. A phase-field investigation into mushy zone constant impacts on PCM melting: from interface morphology to convective heat transfer. *Int Commun Heat Mass Transfer*. 2026;172:110402. doi:10.1016/j.icheatmasstransfer.2025.110402
31. Pan C, Charles J, Vermaak N, et al. Experimental, numerical and analytic study of unconstrained melting in a vertical cylinder with a focus on mushy region effects. *Int J Heat Mass Transf*. 2018;124:1015-1024. doi:10.1016/j.ijheatmasstransfer.2018.04.009
32. Doyle L, García-Molleja J, González C. Low-velocity penetration impact behavior of triply periodic minimal surface strut-based lattices. *Adv Eng Mater*. 2025;27(10):2402999. doi:10.1002/adem.202402999
33. Junaid M, Saha G, Shahrear P, Saha SC. Phase change material performance in chamfered dual enclosures: exploring the roles of geometry, inclination angles and heat flux. *Int J Thermofluids*. 2024;24:100919. doi:10.1016/j.ijft.2024.100919
34. Chtourou S, Djemel H, Kaffel M, et al. A CFD investigation of a turbulent flow in a corrugated plate heat exchanger. In: Kharrat M, Baccar M, Dammak F, eds. *Advances in Mechanical Engineering, Materials and Mechanics*. Springer International Publishing; 2021:179-186. doi:10.1007/978-3-030-52071-7_25
35. Djemel H, Chtourou S, Kaffel M, Baccar M. Thermal analysis of a plate heat exchanger (PHE) fitted with carding tool patterns using CFD modeling. *Heat Trans Res*. 2024;55(6):19-45. doi:10.1615/HeatTransRes.2023048050
36. Arqam M, Raffa LS, Ryall M, Islam MS, Bennett NS. Numerical and experimental analysis of triply periodic minimal surface (TPMS)-based metal lattice heat sinks integrated with different phase change materials for enhanced thermal management of electronics. *J Energy Storage*. 2025;132:117784.
37. Su H, Liu F, Zhang Z, et al. Stage-dependent performance and design criteria of functionally graded TPMS–PCM composites: joint effects of gradient function and steepness. *Int Commun Heat Mass Transfer*. 2026;174:110913. doi:10.1016/j.icheatmasstransfer.2026.110913
38. Zhang X, Yan G, Cheng W, et al. Topology optimization for accelerating thermal charging in TPMS-based latent heat storage units: a rapid method for determining porosity distribution. *Energy*. 2026;351:140881. doi:10.1016/j.energy.2026.140881

**How to cite this article:** Chtourou S, Ali IB, Chelbi L, Baccar M. Topology-porosity assessment of TPMS-based aluminum foams for accelerated melting of RT31 phase change material. *Environ Prog Sustainable Energy*. 2026;e70734. doi:10.1002/ep.70734

19447450, 0, Downloaded from https://aiche.onlinelibrary.wiley.com/doi/10.1002/ep.70734 by Mahidol University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License